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Unit-3

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- Basic Machine Learning Algorithms
- Association Rule mining
- Linear Regression
- Logistic Regression
- Classifiers – k-Nearest Neighbors (k-NN) k-means
- Decision tree – Naive Bayes- Ensemble Methods – Random Forest.
- Feature Generation and Feature Selection –
- Feature Selection algorithms – Filters; Wrappers;
- Decision Trees; Random Forests.

Machine Learning

“Learning denotes changes in a system that ... enable a system to do the same task more efficiently the next time.” –Herbert Simon

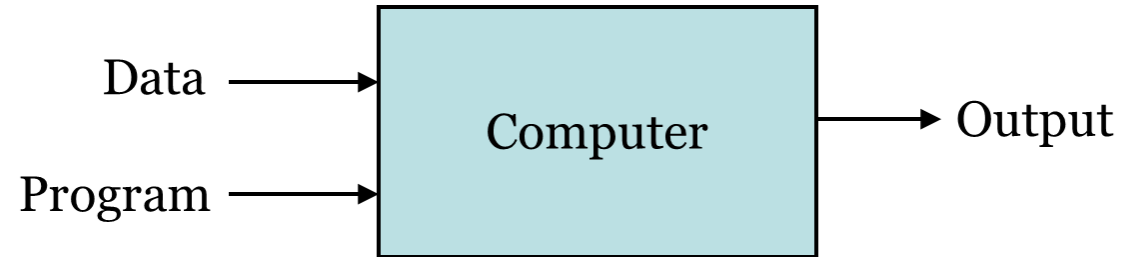
“Learning is constructing or modifying representations of what is being experienced.”
–Ryszard Michalski

“Learning is making useful changes in our minds.” –Marvin Minsky

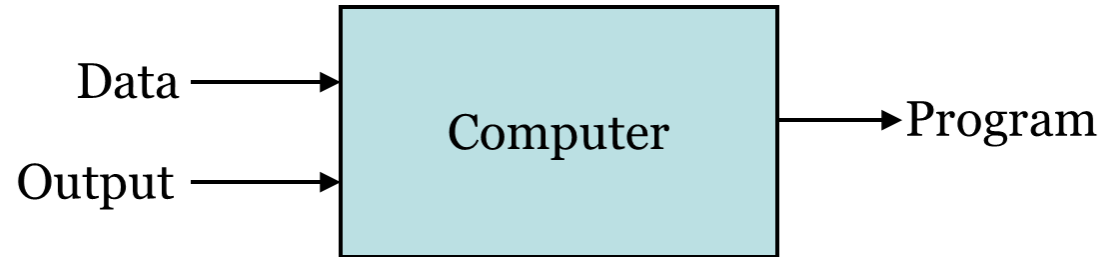
Machine Learning

- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck
- Let the data do the work instead!

Traditional Programming



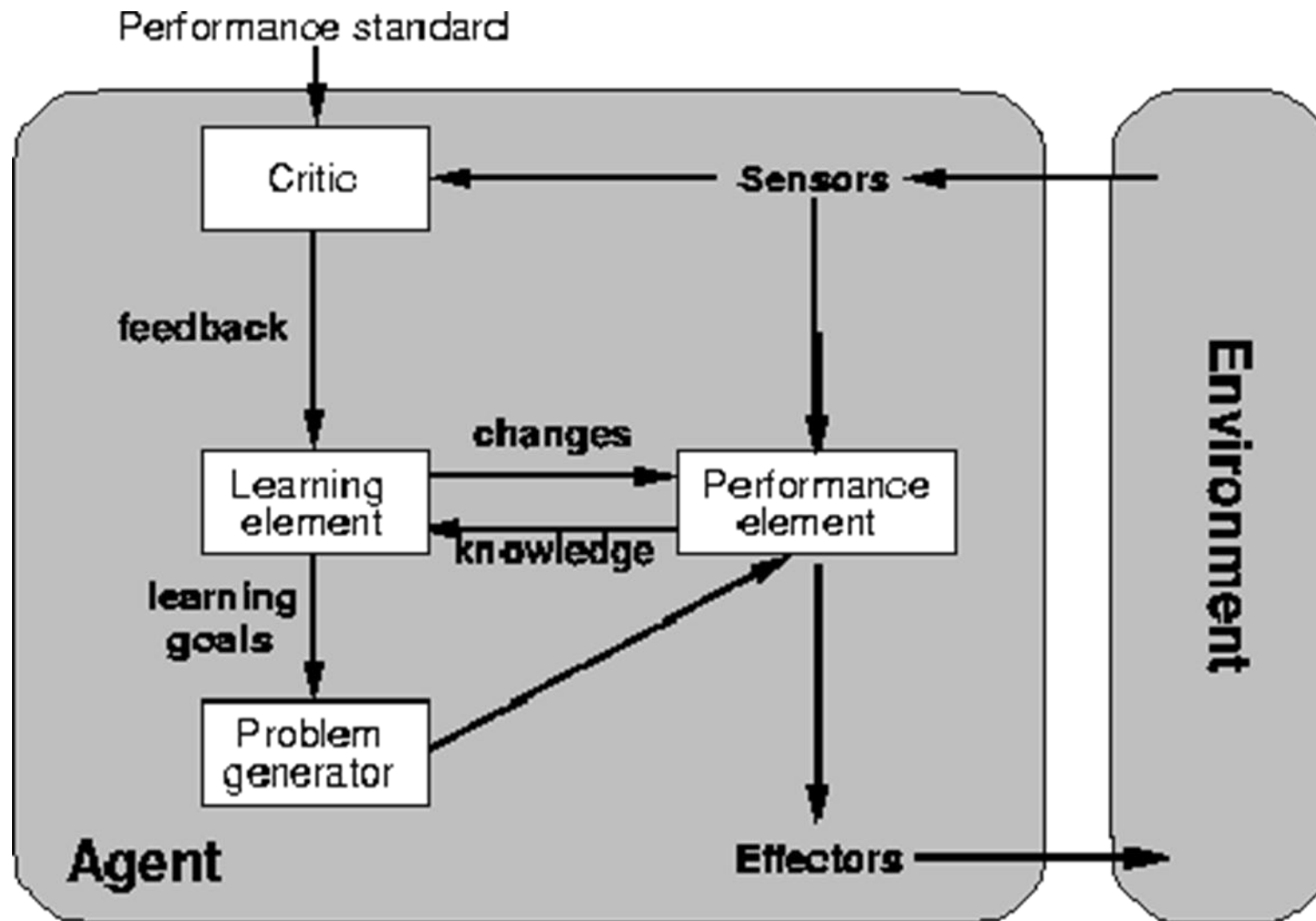
Machine Learning



Machine Learning

- Understand and improve efficiency of human learning
 - Use to improve methods for teaching and tutoring people (e.g., better computer-aided instruction)
- Discover new things or structure that were previously unknown to humans
 - Examples: data mining, scientific discovery
- Fill in skeletal or incomplete specifications about a domain
 - Large, complex AI systems cannot be completely derived by hand and require dynamic updating to incorporate new information.
 - Learning new characteristics expands the domain or expertise and lessens the “brittleness” of the system
- Build software agents that can adapt to their users or to other software agents

A general model of learning agents



Major paradigms of machine learning

- **Rote learning** - One-to-one mapping from inputs to stored representation. “Learning by memorization.” Association-based storage and retrieval.
- **Induction** - Use specific examples to reach general conclusions
- **Clustering** - Unsupervised identification of natural groups in data
- **Analogy** - Determine correspondence between two different representations
- **Discovery** - Unsupervised, specific goal not given
- **Genetic algorithms** - “Evolutionary” search techniques, based on an analogy to “survival of the fittest”
- **Reinforcement** - Feedback (positive or negative reward) given at the end of a sequence of steps

Machine Learning in practice

- Understanding domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learning models
- Interpreting results
- Consolidating and deploying discovered knowledge
- Loop

Association Rule Mining

- Association rule mining is a technique used to uncover hidden relationships between variables in large datasets.
- Association Rule Mining is a method for identifying frequent patterns, correlations, associations, or causal structures in data sets found in numerous databases such as relational databases, transactional databases, and other types of data repositories.
- Given a set of transactions, the goal of association rule mining is to find the rules that allow us to predict the occurrence of a specific item based on the occurrences of the other items in the transaction.
- An association rule consists of two parts:
 - an Antecedent (if) and
 - a Consequent (then)

Association Rule Mining

- An antecedent is something found in data, and a consequent is something located in conjunction with the antecedent.
- Consider the following association rule:

“If a customer buys bread, he’s 70% likely of buying milk.”

Bread is the antecedent in the given association rule, and milk is the consequent.

Metrics for evaluating association rules

- Association rules are carefully derived from the dataset. Several metrics are commonly used to evaluate the performance of association rule mining algorithms. Let us consider the following transaction table.

Transaction ID	Items purchased
1	Item1, Item2
2	Item1, Item3 ,Item4, Item5
3	Item2, Item3, Item4, Item6
4	Item1, Item2, Item3, Item4
5	Item1, Item2, Item3, Item6

Metrics for evaluating association rules

Support

It is the proportion of transactions in the dataset that contain a specific itemset. It indicates the frequency with which the itemset appears in the data. Higher support values indicate that the rule is more common or significant in the dataset. Rules with low support are considered less relevant.

Transaction ID	Items purchased
1	Item1, Item2
2	Item1, Item3 ,Item4, Item5
3	Item2, Item3, Item4, Item6
4	Item1, Item2, Item3, Item4
5	Item1, Item2, Item3, Item6

There are five transactions; three of those have Item 4 appearing in them.

$$\text{Support}\{Item\ 4\} = \frac{3}{5}$$

Out of five transactions, {Item 2, Item 6} appears together in two transactions

$$\text{Support}\{Item\ 2, Item\ 6\} = \frac{2}{5}$$

Confidence

It is the ratio of the number of transactions containing both the antecedent and the consequent to the number of transactions containing only the antecedent.

$$\text{Confidence}\{X \rightarrow Y\} = \frac{\text{Support}\{X \rightarrow Y\}}{\text{Support}\{X\}}$$

$$\text{Confidence}\{\text{Item 2} \rightarrow \text{Item 6}\} = \frac{\text{Support}\{\text{Item 2} \rightarrow \text{Item 6}\}}{\text{Support}\{\text{Item 2}\}}$$

$$\text{Confidence}\{\text{Item 2} \rightarrow \text{Item 6}\} = \frac{2/5}{4/5} = 0.5$$

It is not symmetric, meaning the confidence for $\{X \rightarrow Y\}$ is not the same as $\{Y \rightarrow X\}$

Lift

Lift quantifies how likely the consequent is to occur when the antecedent is present compared to when the two events are independent.

$$\begin{aligned} \text{Lift}\{X \rightarrow Y\} &= \frac{\text{Support}\{X \rightarrow Y\}}{\text{Support}\{X\} \times \text{Support}\{Y\}} \\ &= \frac{\text{Confidence}\{X \rightarrow Y\}}{\text{Support}\{Y\}} \end{aligned}$$

$$\text{Lift}\{\text{Item 2} \rightarrow \text{Item 6}\} = \frac{\text{Confidence}\{\text{Item 2} \rightarrow \text{Item 6}\}}{\text{Support}\{\text{Item 6}\}}$$

$$= \frac{0.5}{2/5} = 1.25$$

Use Cases of Association Rule Mining

Association rule mining is commonly used in a variety of applications, some common ones are:

1. Market Basket Analysis

One of the most well-known applications of association rule mining is in market basket analysis. This involves analyzing the items customers purchase together to understand their purchasing habits and preferences.

For example, a retailer might use association rule mining to discover that customers who purchase diapers are also likely to purchase baby formula. We can use this information to optimize product placements and promotions to increase sales.

2. Fraud Detection

We can also use association rule mining to detect fraudulent activity. For example, a credit card company might use association rule mining to identify patterns of fraudulent transactions, such as multiple purchases from the same merchant within a short period of time.

We can then use this information to flag potentially fraudulent activity and take preventative measures to protect customers.

3. Customer Segmentation

Association rule mining can also be used to segment customers based on their purchasing habits. For example, a company might use association rule mining to discover that customers who purchase certain types of products are more likely to be younger. Similarly, they could learn that customers who purchase certain combinations of products are more likely to be located in specific geographic regions.

We can use this information to tailor marketing campaigns and personalized recommendations to specific customer segments.

4. Social network analysis

Various companies use association rule mining to identify patterns in social media data that can inform the analysis of social networks.

For example, an analysis of Twitter data might reveal that users who tweet about a particular topic are also likely to tweet about other related topics, which could inform the identification of groups or communities within the network.

5. Recommendation systems

Association rule mining can be used to suggest items that a customer might be interested in based on their past purchases or browsing history. For example, a music streaming service might use association rule mining to recommend new artists or albums to a user based on their listening history.

Apriori Algorithm

- The Apriori algorithm is a widely used algorithm to find frequent itemsets in a dataset.
- The Apriori algorithm is used to implement **Frequent Pattern Mining (FPM)**. Frequent pattern mining is a data mining technique to discover frequent patterns or relationships between items in a dataset.
- Frequent pattern mining involves finding sets of items or itemsets that occur together frequently in a dataset. These sets of items or itemsets are called *frequent patterns*, and their frequency is measured by the number of transactions in which they occur.
- An itemset is a collection of one or more items that appear together in a transaction or dataset. An itemset can be either a single item, also known as a 1-itemset, or a set of k items, also known as a k -itemset.

- For example, in sales transactions of a retail store, an itemset can be referred to as products purchased together, such as bread and milk, which would be a 2-item set.
- The Apriori algorithm can be used to discover frequent itemsets in the sales transactions of a retail store. For instance, the algorithm might discover that customers who purchase bread and milk together often also purchase eggs. This information can be used to recommend eggs to customers who purchase bread and milk in the future.
- The Apriori algorithm is called "apriori" because it uses prior knowledge about the frequent itemsets. The algorithm uses the concept of "apriori property," which states that if an itemset is frequent, then all of its subsets must also be frequent.

Apriori Algorithm: Example

In this example, we will use a **minimum support threshold** of 3. This means an item set must appear in at least three transactions to be considered frequent.

TID	Items
T1	{milk, bread}
T2	{bread, sugar}
T3	{bread, butter}
T4	{milk, bread, sugar}
T5	{milk, bread, butter}
T6	{milk, bread, butter}
T7	{milk, sugar}
T8	{milk, sugar}
T9	{sugar, butter}
T10	{milk, sugar, butter}
T11	{milk, bread, butter}



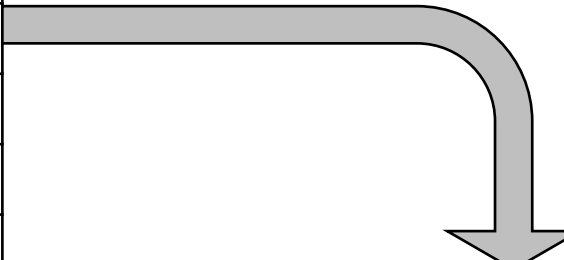
Item	Support (Frequency)
milk	8
bread	7
sugar	5
butter	7

Support for all items is greater than 3. It means that all items are considered as frequent 1-itemsets and will be used to generate candidates for 2-itemsets.

All candidates generated from frequent 1-itemsets identified from the previous step and their support value.

Candidate Item Sets	Support (Frequency)
{milk, bread}	5
{milk, sugar}	3
{milk, butter}	5
{bread, sugar}	2
{bread, butter}	3
{sugar, butter}	2

Now remove candidate item sets that do not meet the minimum support threshold of 3. After this step, frequent 2-itemsets would be - {milk, bread}, {milk, sugar}, {milk, butter}, and {bread, butter}



Candidate Item Sets	Support (Frequency)
{milk, bread}	5
{milk, sugar}	3
{milk, butter}	5
{bread, butter}	3

let's generate candidates for 3-itemsets and calculate their respective support values

Candidate Item Sets	Support (Frequency)
{milk, bread, sugar}	1
{milk, bread, butter}	3
{milk, sugar, butter}	1

As we can see, only one candidate itemset exceeds the minimum defined support threshold –
{milk, bread, butter}

As there is only one 3-itemset exceeding minimum support, we can't generate candidates for 4-itemsets.

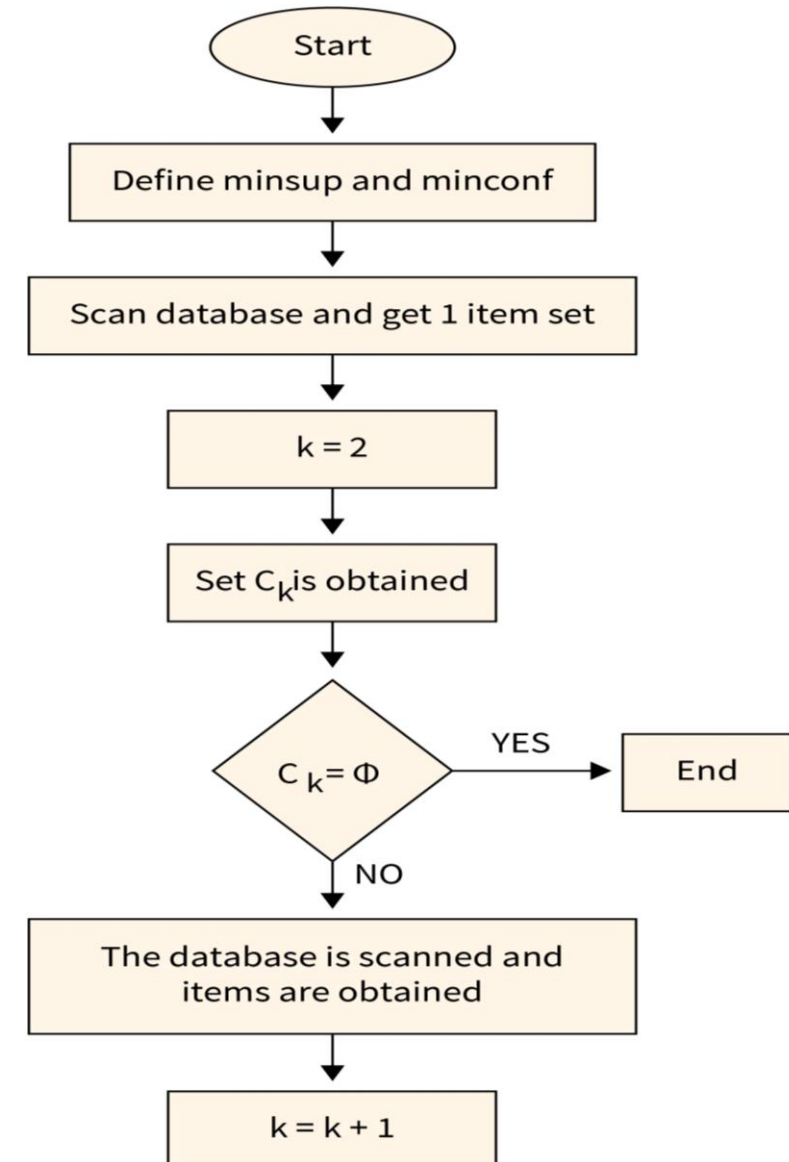
Candidate Item Sets	Support (Frequency)
{milk, bread, butter}	3

we can write the association rules and their respective metrics,

Candidate Item Sets	<i>Support</i>{X → Y}	Support {X}	Confidence
{milk, bread} → butter	3	5	60%
{bread, butter} → milk	3	3	100%
{milk, butter} → bread	3	5	60%

$$\text{Confidence}\{X \rightarrow Y\} = \frac{\text{Support}\{X \rightarrow Y\}}{\text{Support}\{X\}}$$

Apriori Algorithm

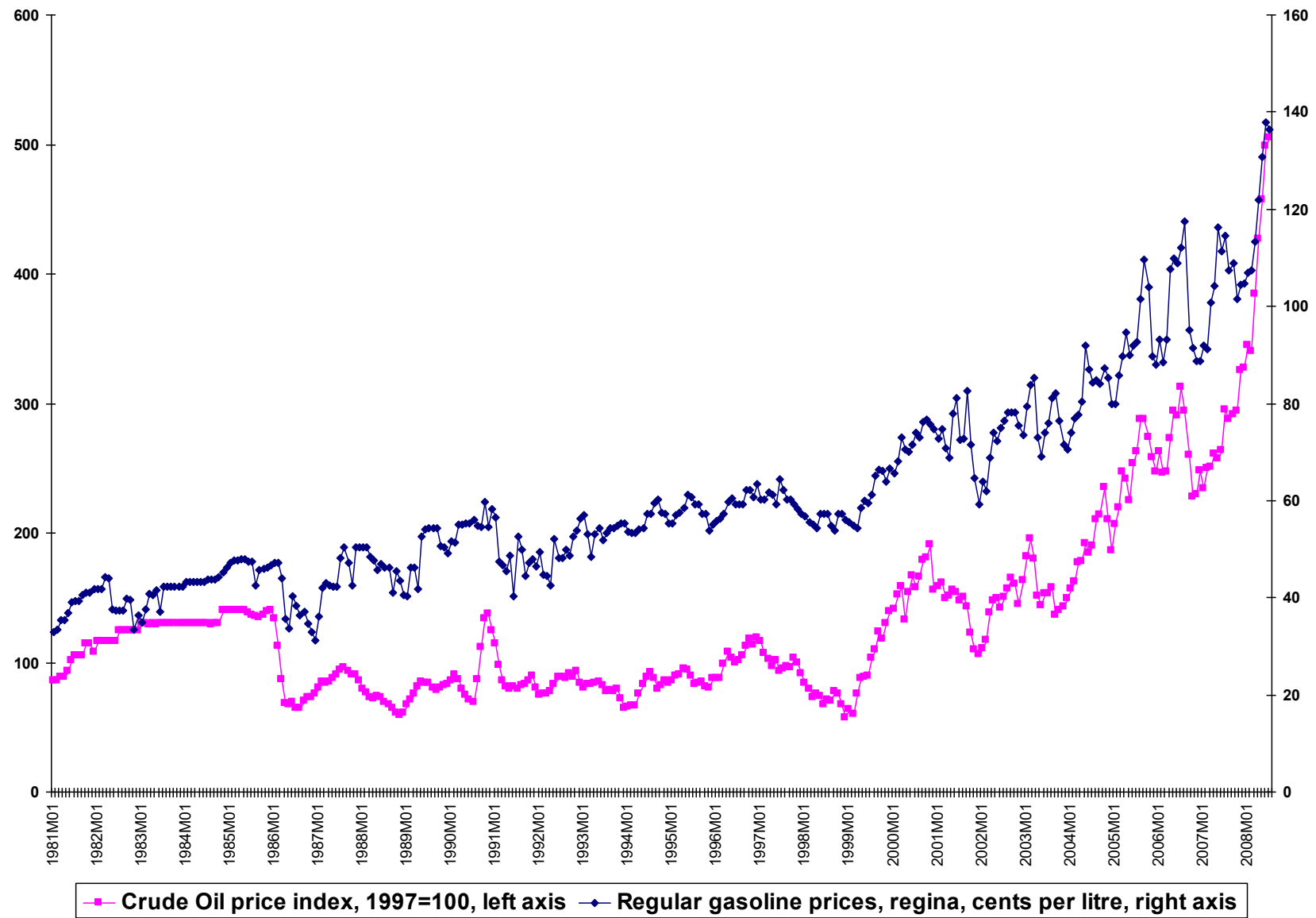


Correlation and Regression

- Relation between variables where changes in some variables may “explain” or possibly “cause” changes in other variables.
- Explanatory variables are termed the **independent** variables and the variables to be explained are termed the **dependent** variables.
- Regression model estimates the nature of the relationship between the independent and dependent variables.
 - Change in dependent variables that results from changes in independent variables, ie. size of the relationship.
 - Strength of the relationship.
 - Statistical significance of the relationship.

Examples

- Dependent variable is retail price of gasoline – independent variable is the price of crude oil.
- Dependent variable is employment income – independent variables might be hours of work, education, occupation, sex, age, region, years of experience, unionization status, etc.
- Price of a product and quantity produced or sold:
 - Quantity sold affected by price. Dependent variable is quantity of product sold – independent variable is price.
 - Price affected by quantity offered for sale. Dependent variable is price – independent variable is quantity sold.



Bivariate and multivariate models

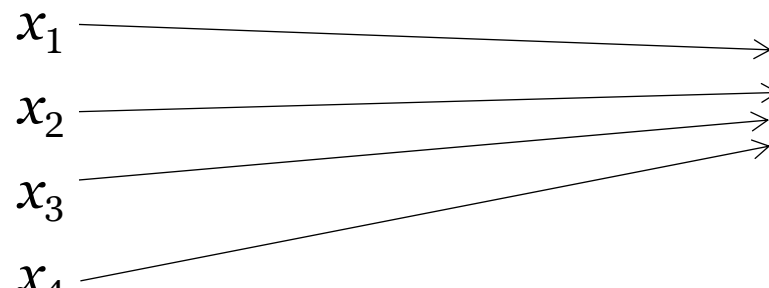
$$100\% Y = 0.2 * x_1 + 0.15 * x_2 + 0.5 * x_3 + 0.15 * x_4$$

Bivariate or simple regression model

(Education) x $\longrightarrow$ y (Income)

Multivariate or multiple regression model

(Education) x_1
(Sex) x_2
(Experience) x_3
(Age) x_4



y (Income)

Model with simultaneous relationship

Price of wheat $\longleftrightarrow$ Quantity of wheat produced

Bivariate or simple linear regression

- x is the independent variable
- y is the dependent variable
- The regression model is

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- The model has two variables, the independent or explanatory variable, x , and the dependent variable y , the variable whose variation is to be explained.
- The relationship between x and y is a linear or straight line relationship.
- Two parameters to estimate – the slope of the line β_1 and the y -intercept β_0 (where the line crosses the vertical axis).
- ε is the unexplained, random, or error component. Much more on this later.

Regression line

- The regression model is $y = \beta_0 + \beta_1 x + \varepsilon$
- Data about x and y are obtained from a sample.
- From the sample of values of x and y , estimates b_0 of β_0 and b_1 of β_1 are obtained using the least squares or another method.
- The resulting estimate of the model is

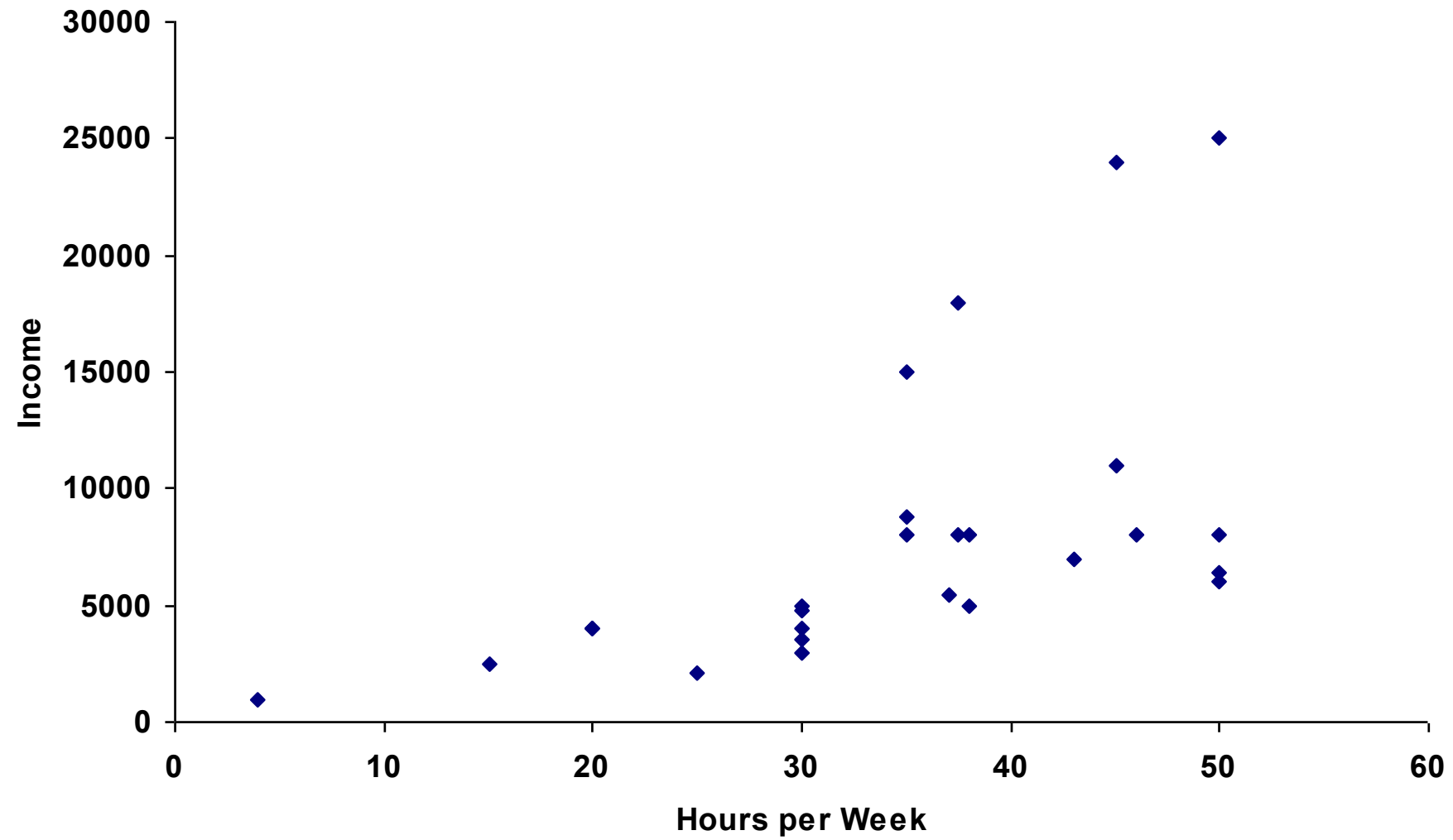
$$\hat{y} = b_0 + b_1 x$$

- The symbol $\hat{y}$ is termed “y hat” and refers to the predicted values of the dependent variable y that are associated with values of x , given the linear model.

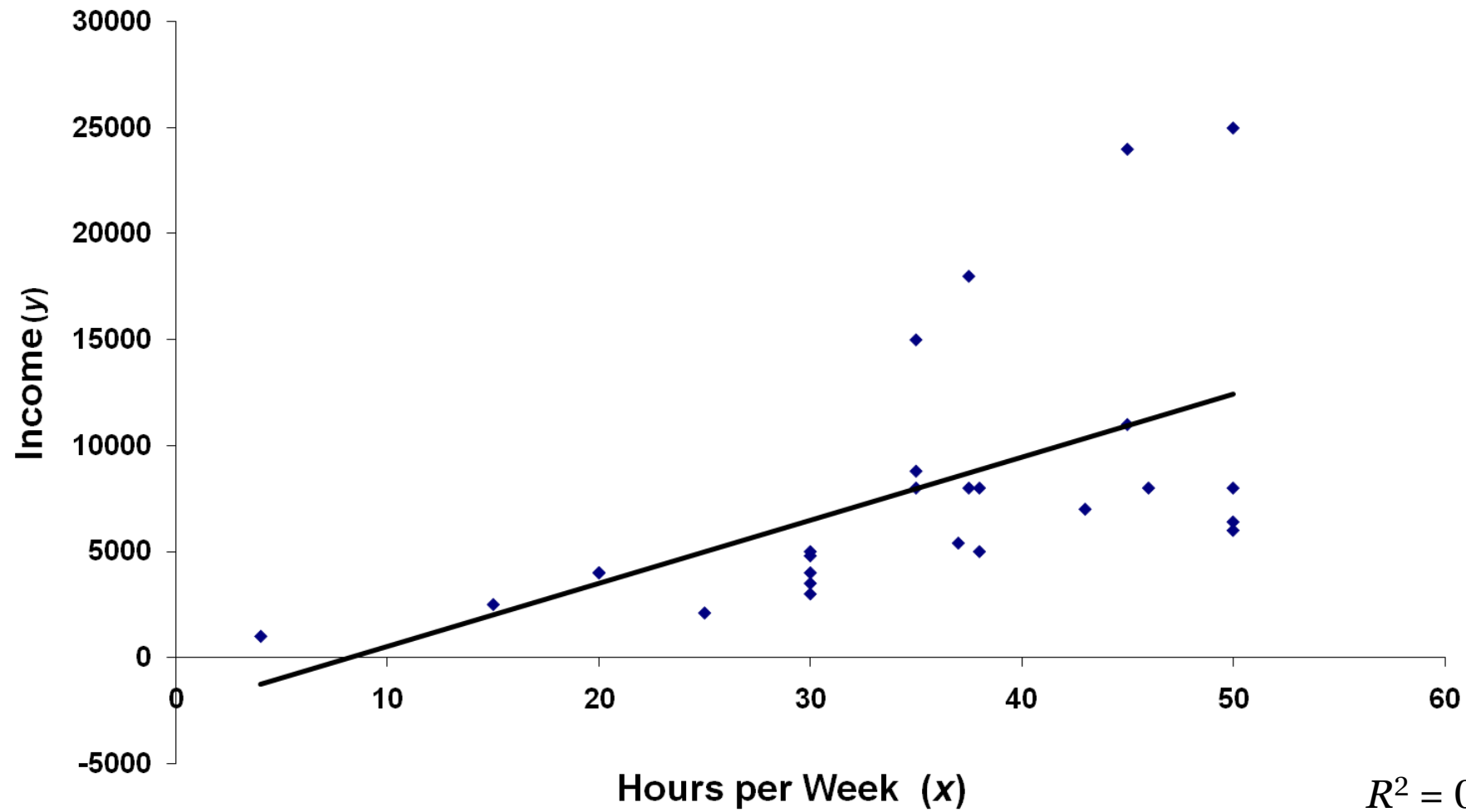
Uses of regression

- Amount of change in a dependent variable that results from changes in the independent variable(s) – can be used to estimate elasticities, returns on investment in human capital, etc.
- Attempt to determine causes of phenomena.
- Prediction and forecasting of sales, economic growth, etc.
- Support or negate theoretical model.
- Modify and improve theoretical models and explanations of phenomena.

Summer Income as a Function of Hours Worked



Summer Income (y) as a Function of Hours Worked (x)



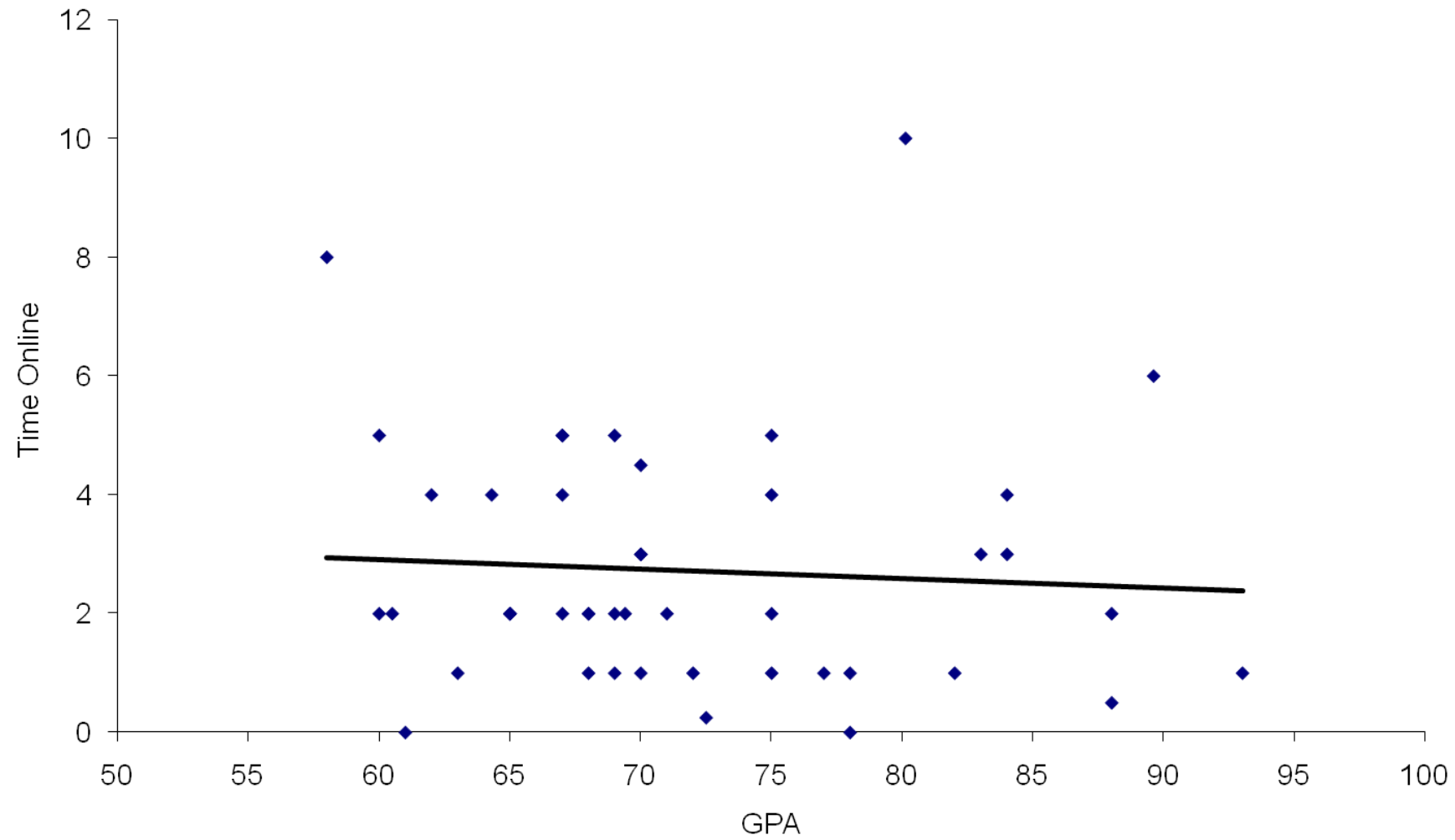
$R^2 = 0.311$
Significance = 0.0031

GPA	Online
75	5
63	1
70	3
68	2
60.5	2
58	8
67	5
80.12	10
64.3	4
84	4
62	4
72.5	0.25
70	4.5
75	2
77	1

GPA	Online
61	0
69	1
84	3
71	2
78	0
75	4
70	3
82	1
60	5
89.6	6
93	1
83	3
65	2
67	4
78	1

GPA	Online
88	0.5
72	1
67	5
69	2
69	5
68	1
65	2
65	2
88	2
68	2
67	2
60	2
70	1
75	1
69.4	2

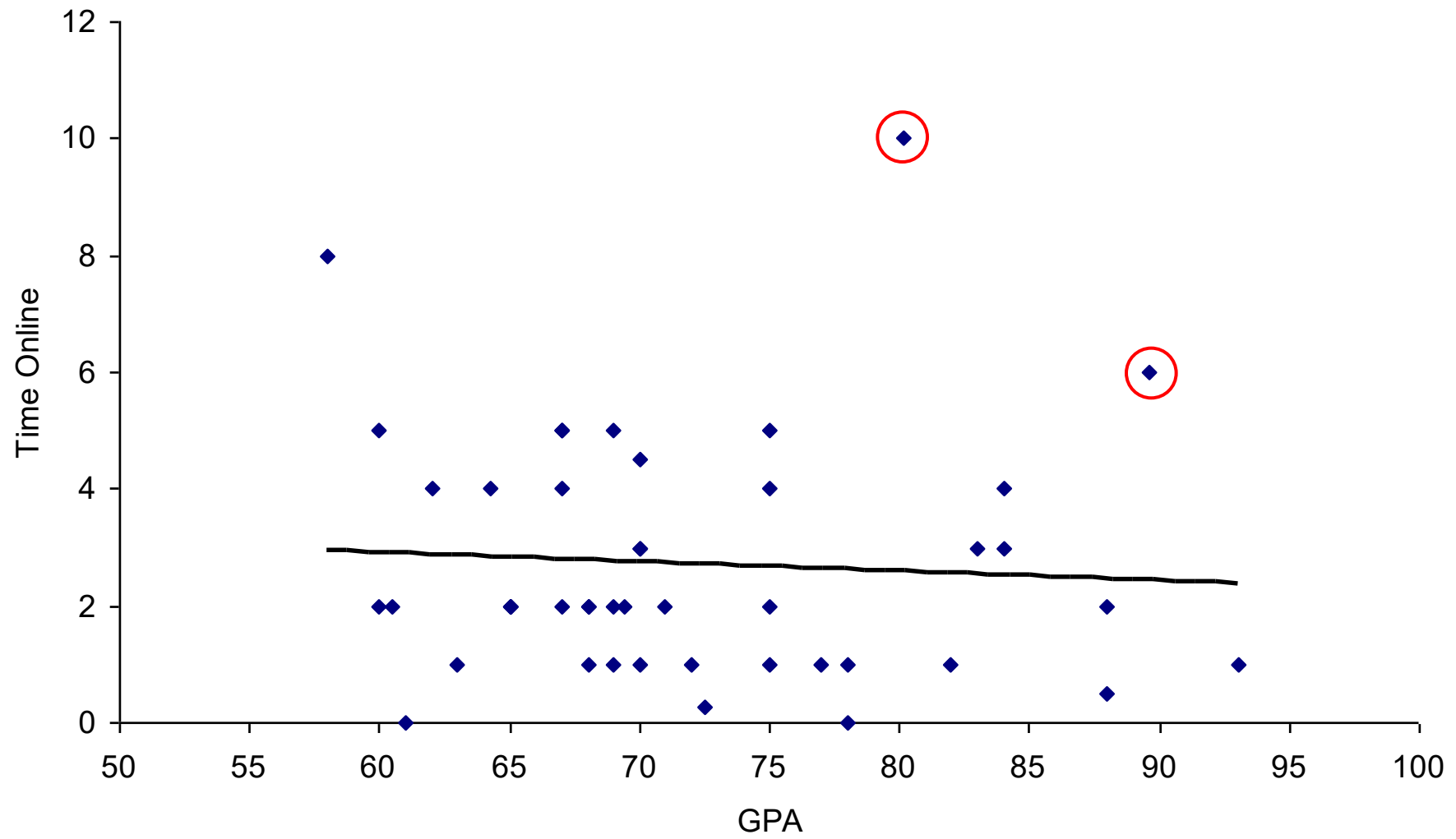
GPA vs. Time Online



Outliers

- Rare, extreme values may distort the outcome.
- Could be an error.
- Could be a very important observation.
- Outlier: more than 3 standard deviations from the mean.

GPA vs. Time Online

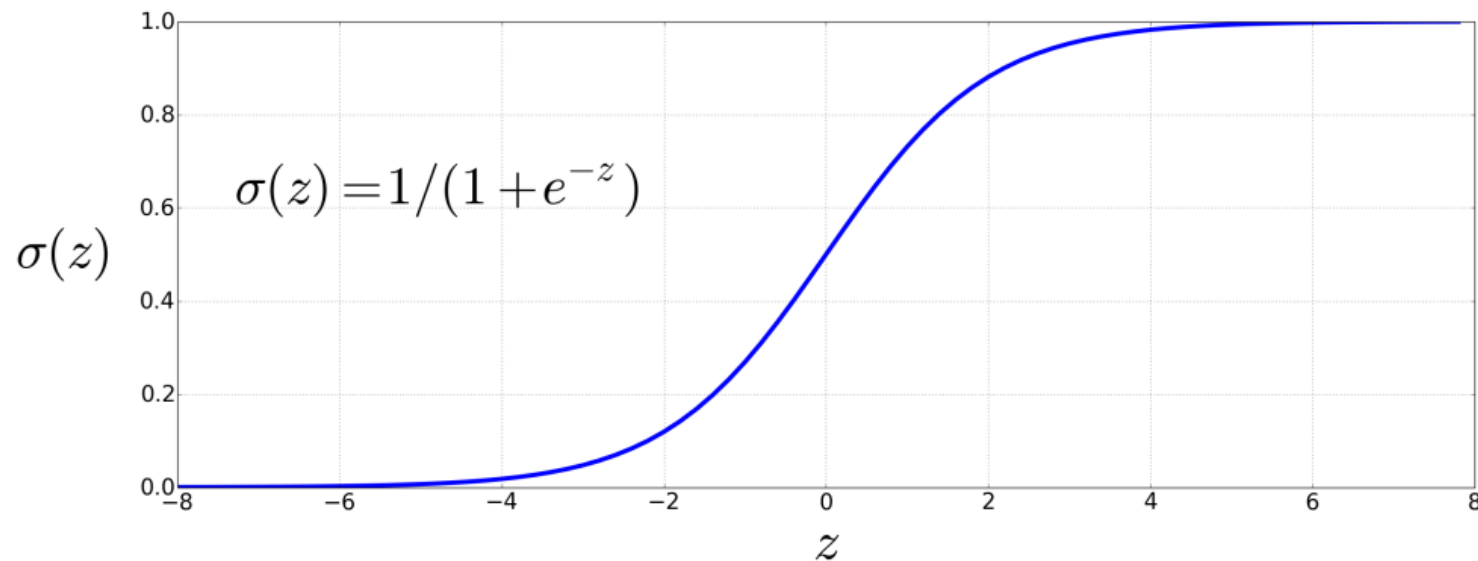


Cost Function

Gradient Descent

Logistic Regression

Instead of fitting a straight line or hyperplane, the logistic regression model uses the logistic function to squeeze the output of a linear equation between 0 and 1. The logistic function is defined as:



The sigmoid function $\sigma(z) = \frac{1}{(1+e^{-z})}$ takes a real value and maps it to the range (0,1). It is nearly linear around 0 but outlier values get squashed toward 0 or 1.

Logistic Regression

The step from linear regression to logistic regression is kind of straightforward. In the linear regression model, we have modelled the relationship between outcome and features with a linear equation:

$$\hat{y}^{(i)} = \beta_0 + \beta_1 x_1^{(i)} + \dots + \beta_p x_p^{(i)}$$

For classification, we prefer probabilities between 0 and 1, so we wrap the right side of the equation into the logistic function. This forces the output to assume only values between 0 and 1.

$$P(y^{(i)} = 1) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_1^{(i)} + \dots + \beta_p x_p^{(i)}))}$$

Logistic Regression

The step from linear regression to logistic regression is kind of straightforward. In the linear regression model, we have modelled the relationship between outcome and features with a linear equation:

$$\hat{y}^{(i)} = \beta_0 + \beta_1 x_1^{(i)} + \dots + \beta_p x_p^{(i)}$$

For classification, we prefer probabilities between 0 and 1, so we wrap the right side of the equation into the logistic function. This forces the output to assume only values between 0 and 1.

$$P(y^{(i)} = 1) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_1^{(i)} + \dots + \beta_p x_p^{(i)}))}$$

Common Distance measures:

Distance measure will determine how the similarity of two elements is calculated.

1. The Euclidean distance
2. Manhattan distance
3. Hamming distance
4. Mahalanobis distance
5. Maximum norm
6. Inner product space

K-Means Clustering

- K-Means clustering intends to partition n objects into k clusters in which each object belongs to the cluster with the nearest mean.
- The best number of clusters k leading to the greatest separation (distance) is not known a priori and must be computed from the data.
- The objective of K-Means clustering is to minimize total intra-cluster variance, or, the squared error function:

Algorithm:

1. Clusters the data into k groups where k is predefined.
2. Select k points at random as cluster centers.
3. Assign objects to their closest cluster center according to the Euclidean distance function.
4. Calculate the centroid or mean of all objects in each cluster.
5. Repeat steps 2, 3 and 4 until the same points are assigned to each cluster in consecutive rounds.

The diagram shows the objective function formula for K-Means clustering: $J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$. Annotations include: 'number of clusters' pointing to k , 'number of cases' pointing to n , 'case i ' pointing to $x_i^{(j)}$, 'centroid for cluster j ' pointing to c_j , and 'Distance function' pointing to the norm $\|x_i^{(j)} - c_j\|^2$. The entire expression is labeled 'objective function' with an arrow pointing to J .

$$\text{objective function} \leftarrow J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$$

K-Means Clustering: Example

Suppose we want to group the visitors to a website using just their age as follows:

15,15,16,19,19,20,20,21,22,28,35,40,41,42,43,44,60,61,65

$$n = 19$$

Initial clusters (random centroid or average):

$$k = 2$$

$$c_1 = 16$$

$$c_2 = 22$$

$$\text{Distance 1} = |x_i - c_1|$$

$$\text{Distance 2} = |x_i - c_2|$$

K-Means Clustering: Example

Iteration 1:

x_i	c_1	c_2	Distance 1	Distance 2	Nearest Cluster	New Centroid
15	16	22	1	7	1	15.33
15	16	22	1	7	1	
16	16	22	0	6	1	
19	16	22	3	3	2	36.25
19	16	22	3	3	2	
20	16	22	4	2	2	
20	16	22	4	2	2	
21	16	22	5	1	2	
22	16	22	6	0	2	
28	16	22	12	6	2	
35	16	22	19	13	2	
40	16	22	24	18	2	
41	16	22	25	19	2	
42	16	22	26	20	2	
43	16	22	27	21	2	
44	16	22	28	22	2	
60	16	22	44	38	2	
61	16	22	45	39	2	
65	16	22	49	43	2	

K-Means Clustering: Example

Iteration 2:

x_i	c_1	c_2	Distance 1	Distance 2	Nearest Cluster	New Centroid
15	15.33	36.25	0.33	21.25	1	18.56
15	15.33	36.25	0.33	21.25	1	
16	15.33	36.25	0.67	20.25	1	
19	15.33	36.25	3.67	17.25	1	
19	15.33	36.25	3.67	17.25	1	
20	15.33	36.25	4.67	16.25	1	
20	15.33	36.25	4.67	16.25	1	
21	15.33	36.25	5.67	15.25	1	
22	15.33	36.25	6.67	14.25	1	
28	15.33	36.25	12.67	8.25	2	45.9
35	15.33	36.25	19.67	1.25	2	
40	15.33	36.25	24.67	3.75	2	
41	15.33	36.25	25.67	4.75	2	
42	15.33	36.25	26.67	5.75	2	
43	15.33	36.25	27.67	6.75	2	
44	15.33	36.25	28.67	7.75	2	
60	15.33	36.25	44.67	23.75	2	
61	15.33	36.25	45.67	24.75	2	
65	15.33	36.25	49.67	28.75	2	

K-Means Clustering: Example

Iteration 3:

x_i	c_1	c_2	Distance 1	Distance 2	Nearest Cluster	New Centroid
15	18.56	45.9	3.56	30.9	1	19.50
15	18.56	45.9	3.56	30.9	1	
16	18.56	45.9	2.56	29.9	1	
19	18.56	45.9	0.44	26.9	1	
19	18.56	45.9	0.44	26.9	1	
20	18.56	45.9	1.44	25.9	1	
20	18.56	45.9	1.44	25.9	1	
21	18.56	45.9	2.44	24.9	1	
22	18.56	45.9	3.44	23.9	1	
28	18.56	45.9	9.44	17.9	1	
35	18.56	45.9	16.44	10.9	2	47.89
40	18.56	45.9	21.44	5.9	2	
41	18.56	45.9	22.44	4.9	2	
42	18.56	45.9	23.44	3.9	2	
43	18.56	45.9	24.44	2.9	2	
44	18.56	45.9	25.44	1.9	2	
60	18.56	45.9	41.44	14.1	2	
61	18.56	45.9	42.44	15.1	2	
65	18.56	45.9	46.44	19.1	2	

K-Means Clustering: Example

Iteration 4:

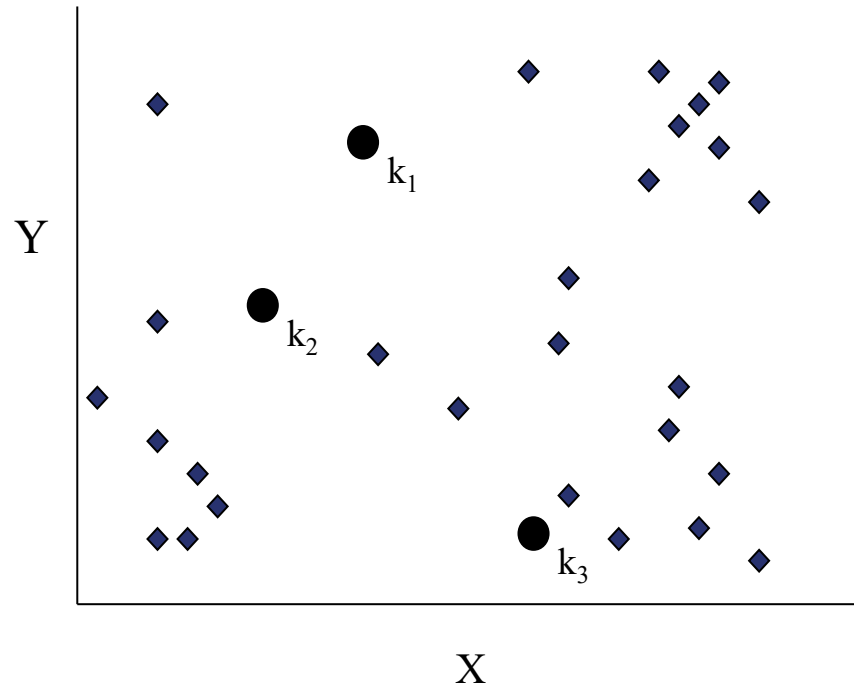
x_i	c_1	c_2	Distance 1	Distance 2	Nearest Cluster	New Centroid
15	19.5	47.89	4.50	32.89	1	19.50
15	19.5	47.89	4.50	32.89	1	
16	19.5	47.89	3.50	31.89	1	
19	19.5	47.89	0.50	28.89	1	
19	19.5	47.89	0.50	28.89	1	
20	19.5	47.89	0.50	27.89	1	
20	19.5	47.89	0.50	27.89	1	
21	19.5	47.89	1.50	26.89	1	
22	19.5	47.89	2.50	25.89	1	
28	19.5	47.89	8.50	19.89	1	
35	19.5	47.89	15.50	12.89	2	47.89
40	19.5	47.89	20.50	7.89	2	
41	19.5	47.89	21.50	6.89	2	
42	19.5	47.89	22.50	5.89	2	
43	19.5	47.89	23.50	4.89	2	
44	19.5	47.89	24.50	3.89	2	
60	19.5	47.89	40.50	12.11	2	
61	19.5	47.89	41.50	13.11	2	
65	19.5	47.89	45.50	17.11	2	

No change between iterations 3 and 4 has been noted.

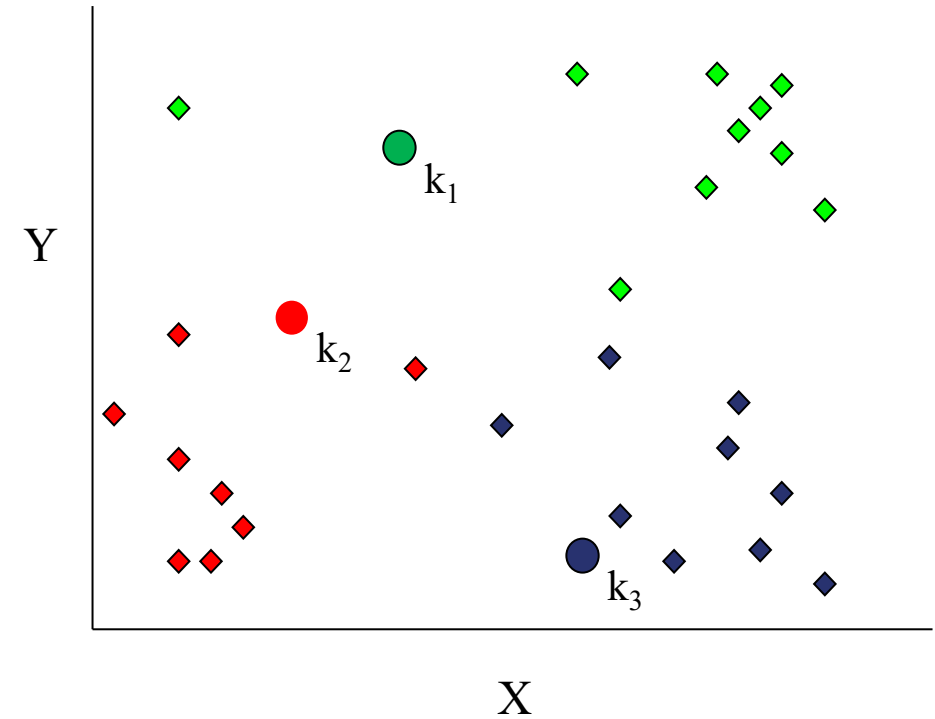
By using clustering, 2 groups have been identified 15-28 and 35-65.

The initial choice of centroids can affect the output clusters, so the algorithm is often run multiple times with different starting conditions in order to get a fair view of what the clusters should be.

K-Means Clustering



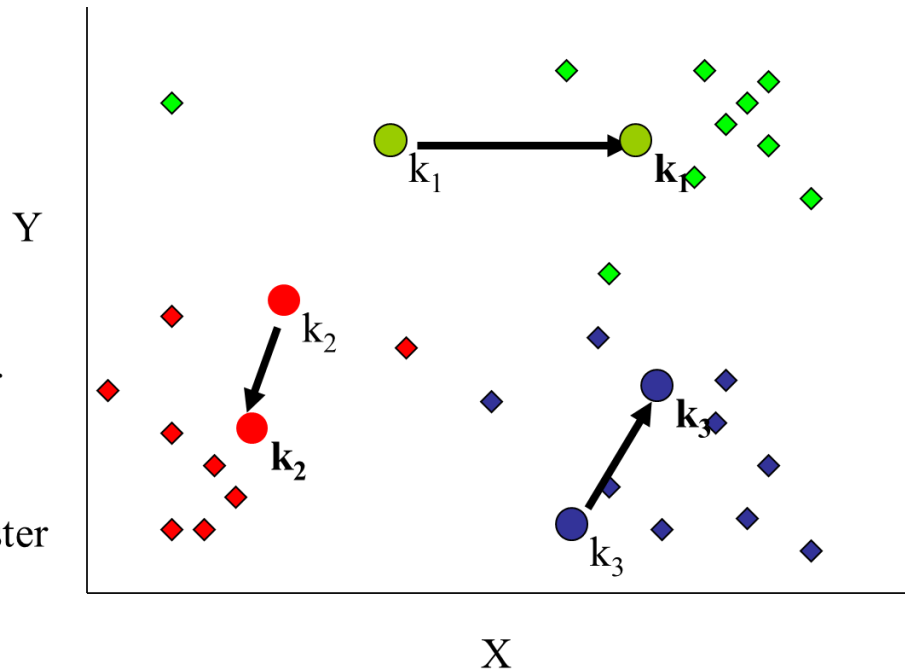
Pick 3 initial cluster centers (randomly)



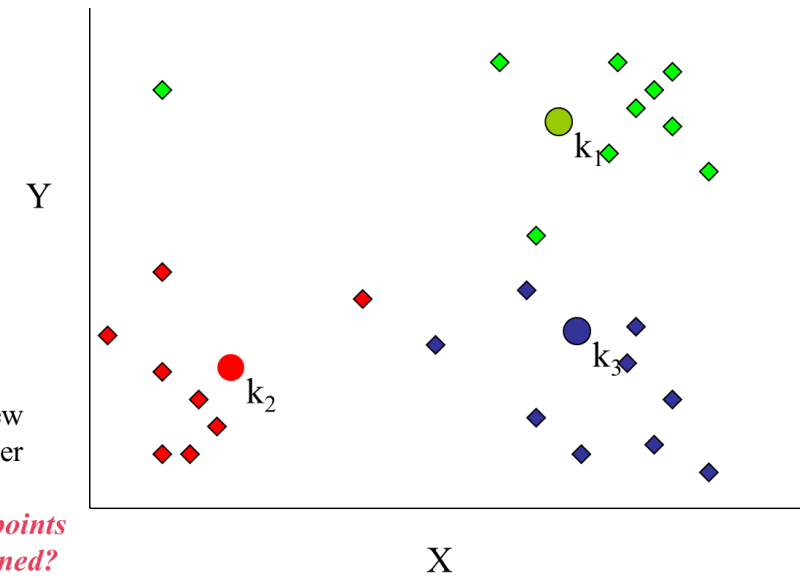
Assign each point to the closest cluster center

K-Means Clustering

Move
each cluster
center
to the mean
of each cluster

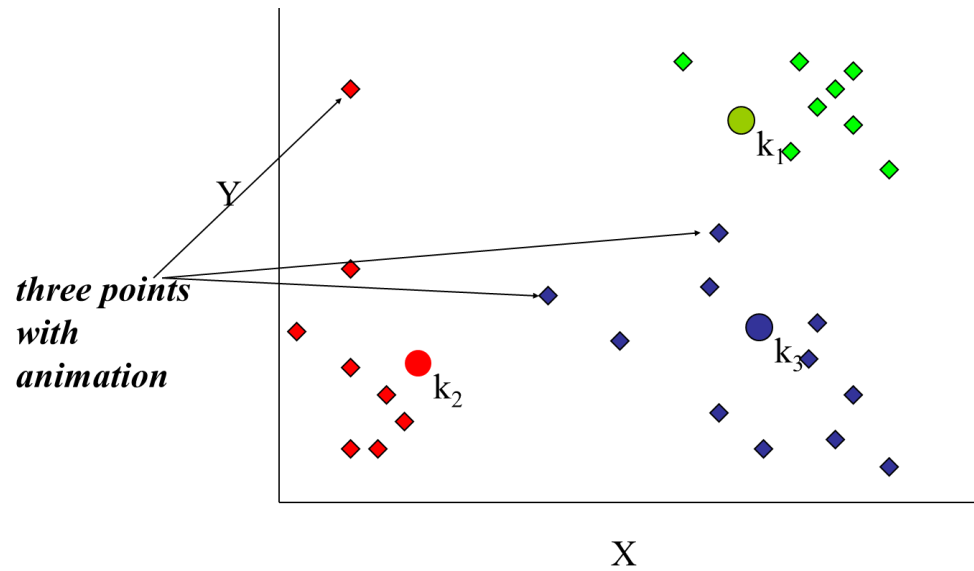


Reassign
points
closest to a
different new
cluster center

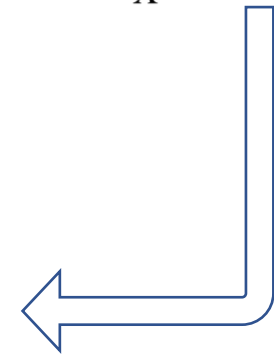
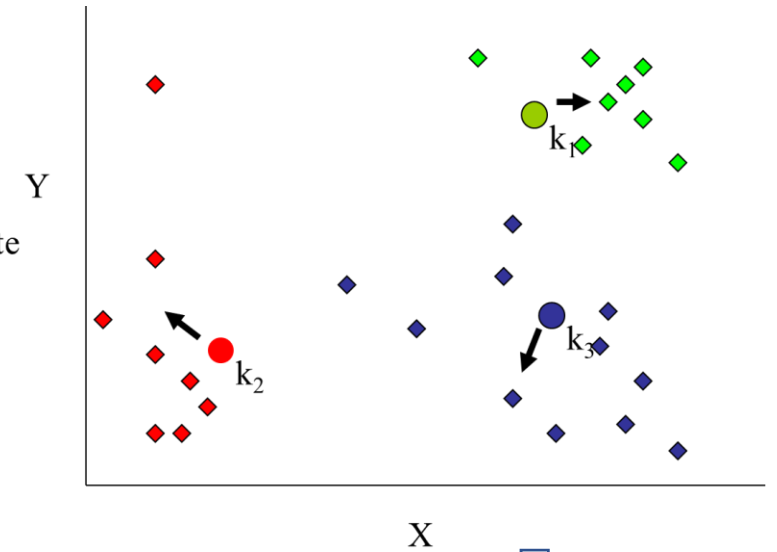


*Q: Which points
are reassigned?*

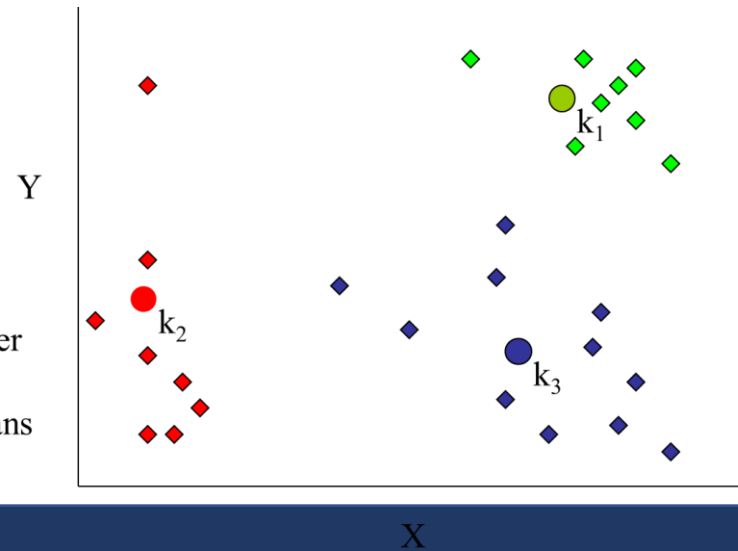
K-Means Clustering



re-compute
cluster
means



move cluster
centers to
cluster means



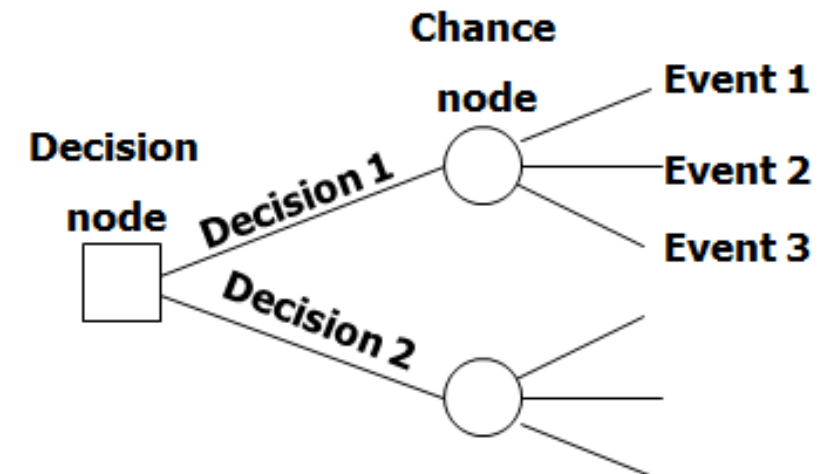
15, 27, 14, 9, 36, 89, 74, 55, 12, 23

Decision Trees

- Decision tree is a simple but powerful learning paradigm. In this method a set of training examples is broken down into smaller and smaller subsets while at the same time an associated decision tree get incrementally developed. At the end of the learning process, a decision tree covering the training set is returned.
- The decision tree can be thought of as a set sentences (in Disjunctive Normal Form) written propositional logic.
- Some characteristics of problems that are well suited to Decision Tree Learning are:
 - Attribute-value paired elements
 - Discrete target function
 - Disjunctive descriptions (of target function)
 - Works well with missing or erroneous training data

Decision Trees

- Classify a pattern through a sequence of questions, next question asked depends on the answer to the current question
- This approach is particularly useful for non-metric data; questions can be asked in a “yes-no” or “true-false” style that do not require any notion of metric
- Sequence of questions is displayed in a directed decision tree
- Root node, links or branches, leaf or terminal nodes
- Classification of a pattern begins at the root node until we reach the leaf node; pattern is assigned the category of the leaf node
- Benefit of decision tree:
 - Interpretability: a tree can be expressed as a logical expression
 - Rapid classification: a sequence of simple queries
 - Higher accuracy & speed



Entropy

- Entropy is a measure of the level of disorder or uncertainty in a given dataset or system. It is a metric that quantifies the amount of information in a dataset
- Entropy is used to determine the best split at each decision node in the tree by calculating the reduction in entropy achieved by splitting the dataset based on a specific feature. The feature with the highest reduction in entropy is chosen as the split point
- a high value of entropy means that the randomness in your system is high, meaning it is difficult to predict the state of atoms or molecules in it. On the other hand, if the entropy is low, predicting that state is much easier.
- The lower this disorder is the more accurate results/predictions you can get.

$$Entropy (S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Information Gain

- Information gain is a measure used in decision trees to determine the usefulness of a feature in classifying a dataset. It is based on the concept of entropy, where entropy is the measure of impurity in a dataset.
- Each decision tree node represents a specific feature, and the branches stemming from that node correspond to the potential values that the feature can take. Information gain is used to determine which feature to split on at each internal node, such that the resulting subsets of data are as pure as possible.
- Information gain is determined by subtracting the entropy of the parent node from the weighted average of the entropies of the child nodes. The formula for information gain can be represented as:

$$\text{Information Gain}(S) = \text{Entropy}(S) - \sum [p(S/A) \cdot \text{Entropy}(S/A)]$$

ID3 Algorithm

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Step 1: Calculate Entropy:

To calculate entropy, we first determine the proportion of positive and negative instances in the dataset:

Positive instances (Play Tennis = Yes): 9

Negative instances (Play Tennis = No): 5

$$Entropy(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

$$Entropy(\text{Play Tennis}) = -(9/14) * \log_2(9/14) - (5/14) * \log_2(5/14) = 0.940$$

Step 2: Calculating Information Gain:

We calculate the information gain for each attribute (Weather, Temperature, Humidity, Windy) and choose the attribute with the highest information gain as the root node.

$$\text{Information Gain}(S) = \text{Entropy}(S) - \sum [p(S/A) \cdot \text{Entropy}(S/A)]$$

Lets consider Wind Attribute first:

$$\begin{aligned} \text{Information Gain}(\text{Play Tennis}, \text{Wind}) = & \text{Entropy}(\text{Play Tennis}) - \\ & [p(\text{Play Tennis} | \text{Wind}=\text{Weak}) \cdot \text{Entropy}(\text{Play Tennis} | \text{Wind}=\text{Weak})] - \\ & [p(\text{Play Tennis} | \text{Wind}=\text{Strong}) \cdot \text{Entropy}(\text{Play Tennis} | \text{Wind}=\text{Strong})] \end{aligned}$$

$$\text{Wind}(\text{Weak}) = 8$$

$$\text{Wind}(\text{Strong}) = 6$$

Wind = 'Weak'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
13	Overcast	Hot	Normal	Weak	Yes

$$\text{Entropy}(\text{Play Tennis} \mid \text{Wind=Weak})] \\ = - p(\text{No}) \cdot \log_2 p(\text{No}) - \\ p(\text{Yes}) \cdot \log_2 p(\text{Yes})$$

$$\text{Entropy}(\text{Play Tennis} \mid \text{Wind=Weak})] \\ = - (2/8) \cdot \log_2(2/8) - \\ (6/8) \cdot \log_2(6/8) \\ = 0.811$$

Wind = 'Strong'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
14	Rain	Mild	High	Strong	No

$$\begin{aligned} \text{Entropy}(\text{Play Tennis} \mid \text{Wind}=\text{Strong})] \\ = - p(\text{No}) \cdot \log_2 p(\text{No}) - \\ p(\text{Yes}) \cdot \log_2 p(\text{Yes}) \end{aligned}$$

$$\begin{aligned} \text{Entropy}(\text{Play Tennis} \mid \text{Wind}=\text{Strong})] \\ = - (3/6) \cdot \log_2(3/6) - \\ (3/6) \cdot \log_2(3/6) \\ = 1 \end{aligned}$$

$$\text{Information Gain(Play Tennis, Wind)} = \text{Entropy(Play Tennis)} - \\ [p(\text{Play Tennis} \mid \text{Wind=Weak}) \cdot \text{Entropy(Play Tennis} \mid \text{Wind=Weak})] - \\ [p(\text{Play Tennis} \mid \text{Wind=Strong}) \cdot \text{Entropy(Play Tennis} \mid \text{Wind=Strong})]$$

$$= 0.940 - [(8/14) \cdot 0.811] - [(6/14) \cdot 1] \\ = 0.048$$

Calculations for wind column is over. Now, we need to apply same calculations for other columns:

Information Gain(Play Tennis, Outlook)

$$\begin{aligned} &= \text{Entropy}(S) - [(5/14) * \text{Entropy}(\text{Sunny}) + (4/14) * \text{Entropy}(\text{Overcast}) + (5/14) * \text{Entropy}(\text{Rainy})] \\ &= 0.246 \end{aligned}$$

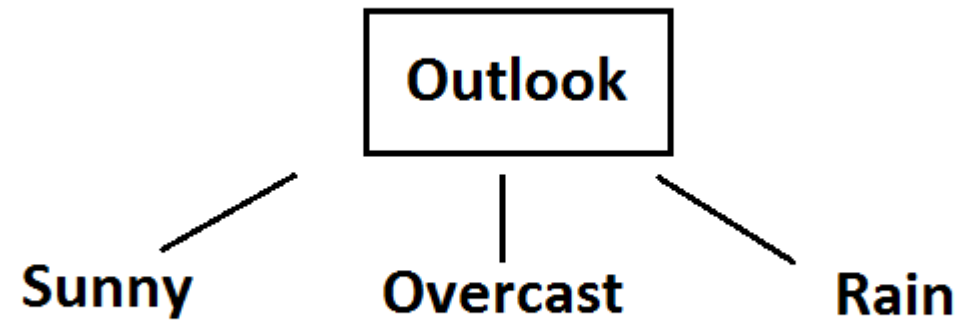
Information Gain(Play Tennis, Temperature)

$$\begin{aligned} &= \text{Entropy}(S) - [(4/14) * \text{Entropy}(\text{Hot}) + (4/14) * \text{Entropy}(\text{Mild}) + (6/14) * \text{Entropy}(\text{Cool})] \\ &= 0.029 \end{aligned}$$

Information Gain(Play Tennis, Humidity)

$$\begin{aligned} &= \text{Entropy}(S) - [(7/14) * \text{Entropy}(\text{High}) + (7/14) * \text{Entropy}(\text{Normal})] \\ &= 0.152 \end{aligned}$$

Outlook factor on decision produces the highest score. That's why, outlook decision will appear in the root node of the tree.



Outlook = 'Sunny'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Here, there are 5 instances for sunny outlook.

$$P(\text{play Tennis} = \text{No}) = 3/5$$

$$P(\text{play Tennis} = \text{Yes}) = 2/5$$

$$\begin{aligned} \text{Information Gain (Outlook = Sunny | } \\ \text{Temperature)} \\ = 0.570 \end{aligned}$$

$$\begin{aligned} \text{Information Gain (Outlook = Sunny | } \\ \text{Humidity)} \\ = 0.970 \end{aligned}$$

$$\begin{aligned} \text{Information Gain (Outlook = Sunny | Wind)} \\ = 0.019 \end{aligned}$$

Now, humidity is the decision because it produces the highest score if outlook were sunny.

Outlook = 'Sunny'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Outlook = 'Overcast'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Outlook = 'Rain'

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Here, there are 5 instances for sunny outlook.

$P(\text{play Tennis} = \text{No}) = ??$

$P(\text{play Tennis} = \text{yes}) = ??$

Information Gain (Outlook = Rain | Temperature)

= 0.019973

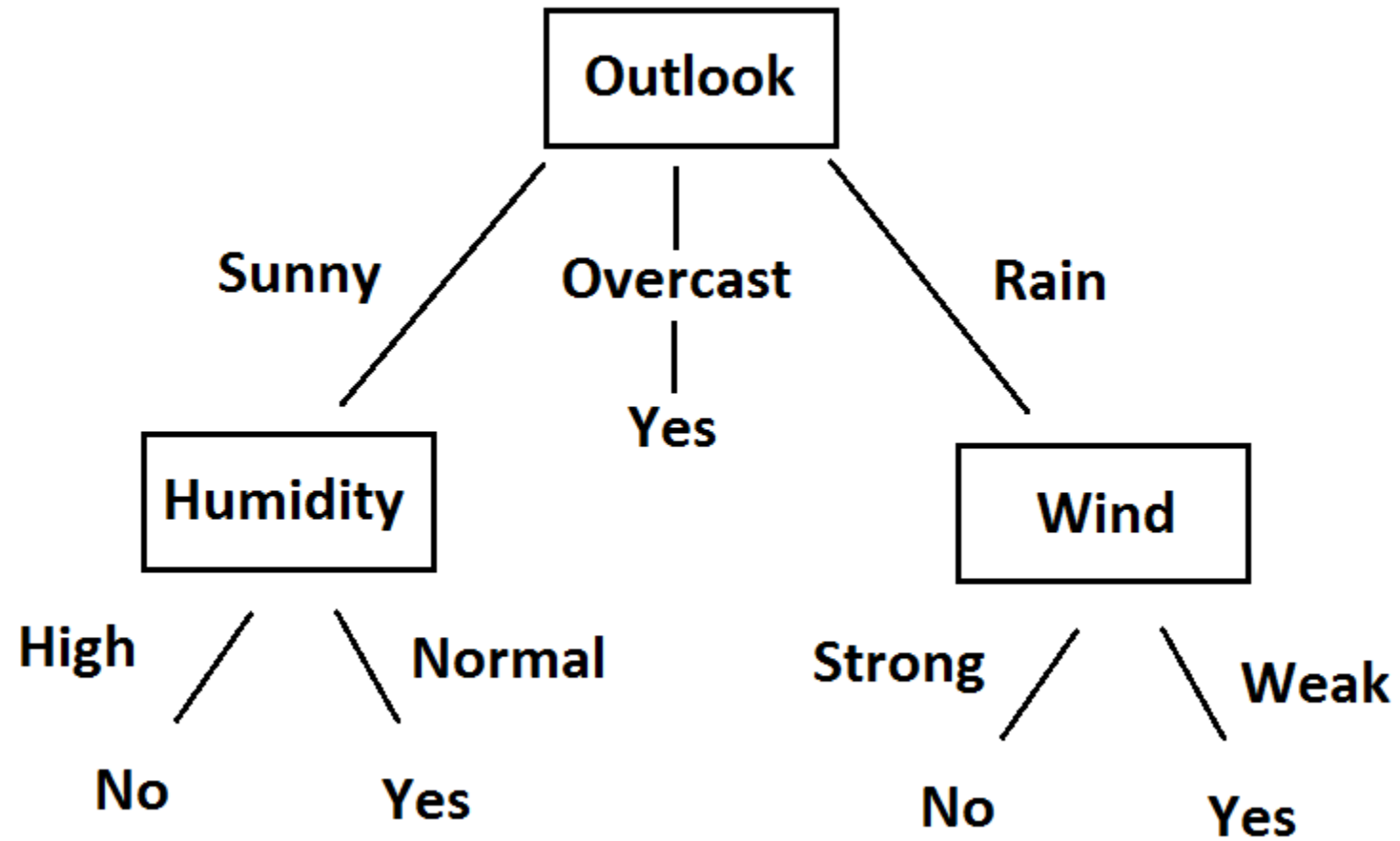
Information Gain (Outlook = Rain | Humidity)

= 0.01997

Information Gain (Outlook = Rain | Wind)

= 0.9709

Here, wind produces the highest score if outlook were rain. That's why, we need to check wind attribute in 2nd level if outlook were rain.



Step 3: Selecting the Best Attribute:

The “Weather” attribute has the highest information gain, so we select it as the root node for our decision tree.

Step 4: Splitting the Dataset:

We split the dataset based on the values of the “Outlook” attribute into three subsets (Sunny, Overcast, Rain).

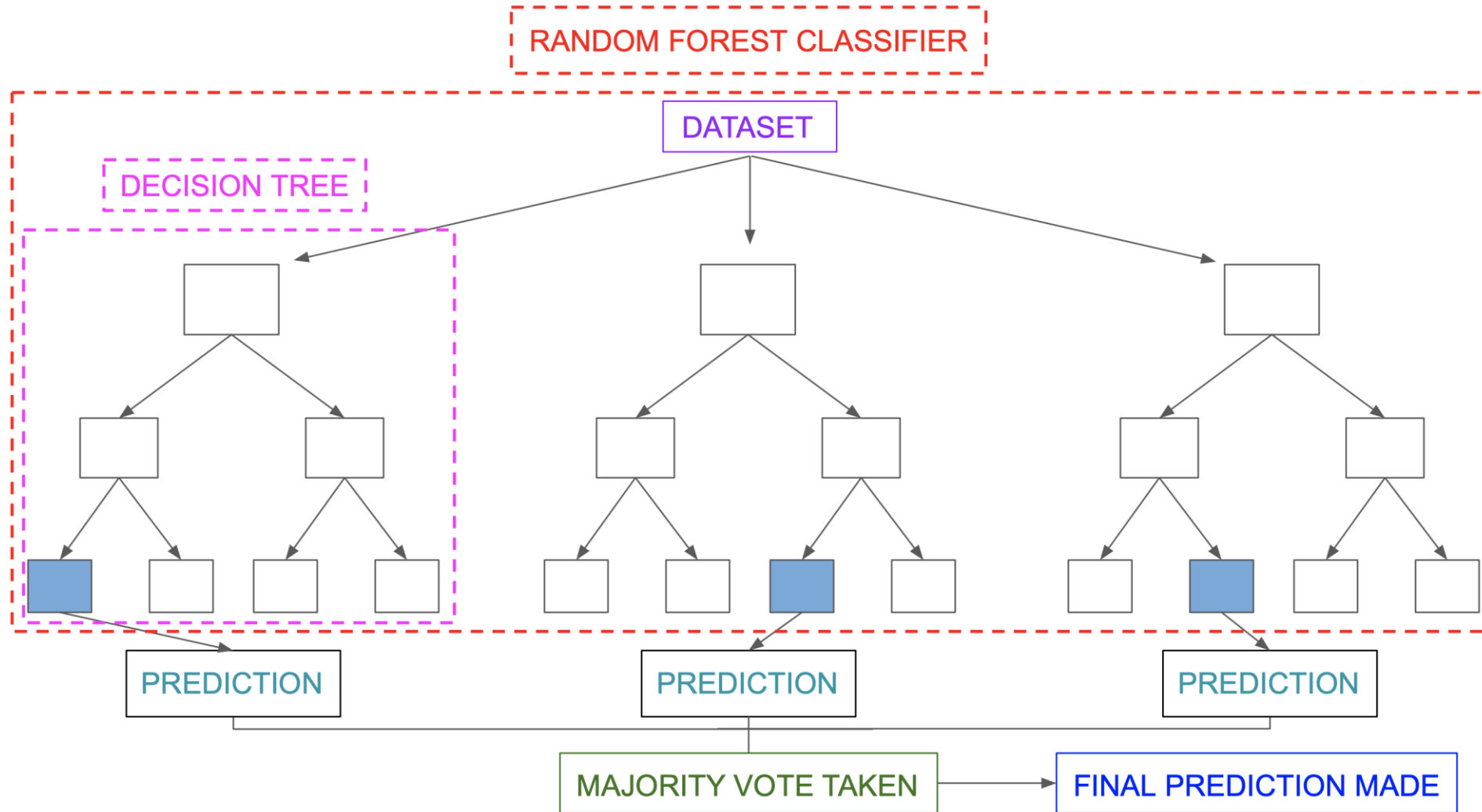
Step 5: Repeat the Process:

Since the “Outlook” attribute has no repeating values in any subset, we stop splitting and label each leaf node with the majority class in that subset. The decision tree will look like below:

Random Forest

- Classification in random forests employs an ensemble methodology to attain the outcome. The training data is fed to train various decision trees. This dataset consists of observations and features that will be selected randomly during the splitting of nodes.
- Every decision tree consists of decision nodes, leaf nodes, and a root node.
- The leaf node of each tree is the final output produced by that specific decision tree.
- The selection of the final output follows the majority-voting system. In this case, the output chosen by the majority of the decision trees becomes the final output of the rain forest system.

Random Forest



Feature Engineering

- A feature (or column) represents a measurable piece of data like name, age or gender. It is the basic building block of a dataset.
- The quality of a feature can vary significantly and has an immense effect on model performance.
- We can improve the quality of a dataset's features in the pre-processing stage using processes like Feature Generation and Feature Selection.
- Feature Generation (also known as feature construction, feature extraction or feature engineering) is the process of transforming features into new features that better relate to the target.
- Feature Generation can improve model performance when there is a feature interaction. Two or more features interact if the combined effect is (greater or less) than the sum of their individual effects.

Feature Engineering

Feature engineering in Machine learning consists of mainly 5 processes:

1. Feature Creation,
2. Feature Transformation,
3. Feature Extraction,
4. Feature Selection, and
5. Feature Scaling.

1. Feature Creation

Feature Creation is the process of generating new features based on domain knowledge or by observing patterns in the data. It is a form of feature engineering that can significantly improve the performance of a machine-learning model.

Types of Feature Creation:

- **Domain-Specific:** Creating new features based on domain knowledge, such as creating features based on business rules or industry standards.
- **Data-Driven:** Creating new features by observing patterns in the data, such as calculating aggregations or creating interaction features.
- **Synthetic:** Generating new features by combining existing features or synthesizing new data points.

2. Feature Transformation

Feature Transformation is the process of transforming the features into a more suitable representation for the machine learning model. This is done to ensure that the model can effectively learn from the data.

Types of Feature Transformation:

- **Normalization:** Rescaling the features to have a similar range, such as between 0 and 1, to prevent some features from dominating others.
- **Scaling:** Rescaling the features to have a similar scale, such as having a standard deviation of 1, to make sure the model considers all features equally.
- **Encoding:** Transforming categorical features into a numerical representation. Examples are one-hot encoding and label encoding.
- **Transformation:** Transforming the features using mathematical operations to change the distribution or scale of the features. Examples are logarithmic, square root, and reciprocal transformations.

3. Feature Extraction

Feature Extraction is the process of creating new features from existing ones to provide more relevant information to the machine learning model. This is done by transforming, combining, or aggregating existing features. E.g. cluster analysis, text analytics, edge detection algorithms, and principal components analysis (PCA).

Types of Feature Extraction:

- **Dimensionality Reduction:** Reducing the number of features by transforming the data into a lower-dimensional space while retaining important information. Examples are PCA and t-SNE.
- **Feature Combination:** Combining two or more existing features to create a new one. For example, the interaction between two features.
- **Feature Aggregation:** Aggregating features to create a new one. For example, calculating the mean, sum, or count of a set of features.
- **Feature Transformation:** Transforming existing features into a new representation. For example, log transformation of a feature with a skewed distribution.

4. Feature Selection

Will Discuss in later slides

5. Feature Scaling

Feature Scaling is the process of transforming the features so that they have a similar scale. This is important in machine learning because the scale of the features can affect the performance of the model.

Types of Feature Scaling:

- **Min-Max Scaling:** Rescaling the features to a specific range, such as between 0 and 1, by subtracting the minimum value and dividing by the range.
- **Standard Scaling:** Rescaling the features to have a mean of 0 and a standard deviation of 1 by subtracting the mean and dividing by the standard deviation.
- **Robust Scaling:** Rescaling the features to be robust to outliers by dividing them by the interquartile range.

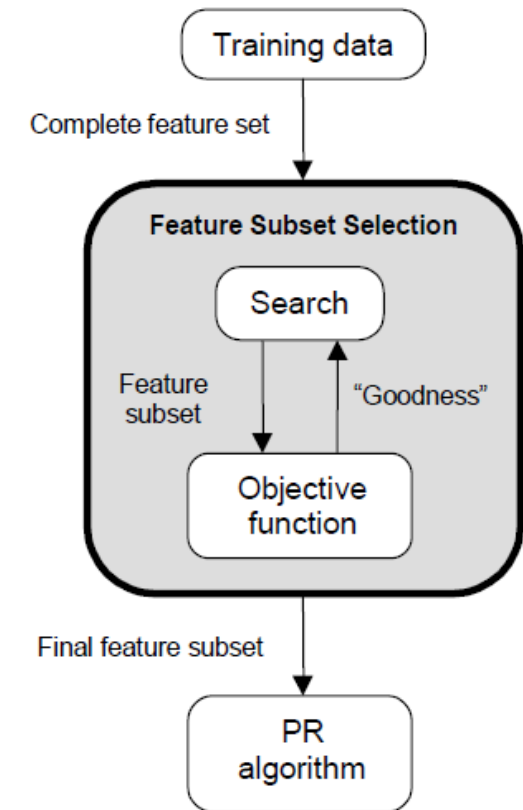
Feature Selection

- Features are one of the salient characteristics of objects through which it can be analyzed.
- An efficient classification scheme depends on the discriminatory and expressive information contained in the features.
- Given a set of N features, the goal of feature selection is to select a subset of K features ($K \ll N$).

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ x_N \end{bmatrix} \rightarrow \mathbf{y} = \begin{bmatrix} x_{i_1} \\ x_{i_2} \\ \cdot \\ \cdot \\ x_{i_K} \end{bmatrix}$$

Feature Selection

- Feature selection is an **optimization** problem.
 - **Step 1:** Search the space of possible feature subsets.
 - **Step 2:** Pick the subset that is **optimal** or **near-optimal** with respect to some **objective** function.



Objectives of Feature Selection

Feature selection has many objectives.

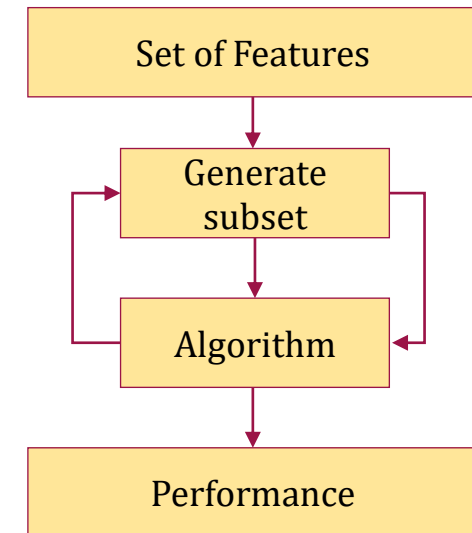
- It eliminates **irrelevant** and **noisy features** by keeping the ones with minimum redundancy and maximum relevance to the target variable.
- It reduces the **computational time** and complexity of training and testing a classifier, so it results in more cost-effective models.
- It improves learning algorithms' performance, **avoids overfitting**, and helps to create better general models.

Types of Feature Selection Methods

- Wrapper Methods
- Filter Methods
- Embedded Methods

Wrapper Methods

- In wrapper methodology, selection of features is done by considering it as a search problem, in which different combinations are made, evaluated, and compared with other combinations. It trains the algorithm by using the subset of features iteratively.
- Some techniques of wrapper methods are:
 - Forward selection
 - Backward elimination
 - Exhaustive Feature Selection
 - Recursive Feature Elimination



Wrapper Methods

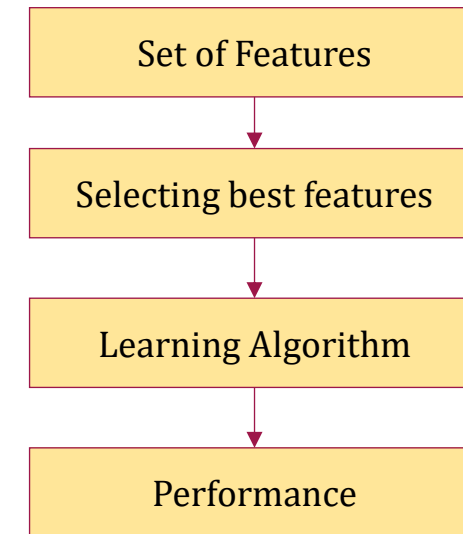
- **Forward selection** - Forward selection is an iterative process, which begins with an empty set of features. After each iteration, it keeps adding on a feature and evaluates the performance to check whether it is improving the performance or not. The process continues until the addition of a new variable/feature does not improve the performance of the model.
- **Backward elimination** - Backward elimination is also an iterative approach, but it is the opposite of forward selection. This technique begins the process by considering all the features and removes the least significant feature. This elimination process continues until removing the features does not improve the performance of the model.

Wrapper Methods

- **Exhaustive Feature Selection**- It evaluates each feature set as brute-force. It means this method tries & make each possible combination of features and return the best performing feature set.
- **Recursive Feature Elimination (RFE)** –RFE is a recursive greedy optimization approach, where features are selected by recursively taking a smaller and smaller subset of features.

Filter Methods

- In Filter Method, features are selected on the basis of statistical measures. This method does not depend on the learning algorithm and chooses the features as a pre-processing step.
- The filter method filters out the irrelevant feature and redundant columns from the model by using different metrics through ranking.
- The advantage of using filter methods is that it needs low computational time and does not overfit the data.
- Some common techniques of Filter methods are as follows:
 - Information Gain
 - Chi-square Test
 - Fisher's Score
 - Missing Value Ratio



Filter Methods

- **Information Gain:** Information gain determines the reduction in entropy while transforming the dataset. It can be used as a feature selection technique by calculating the information gain of each variable with respect to the target variable.
- **Chi-square Test:** Chi-square test is a technique to determine the relationship between the categorical variables. The chi-square value is calculated between each feature and the target variable, and the desired number of features with the best chi-square value is selected.

Filter Methods

- **Fisher's Score:** Fisher's score is one of the popular supervised technique of features selection. It returns the rank of the variable on the fisher's criteria in descending order. Then we can select the variables with a large fisher's score.
- **Missing Value Ratio:** The value of the missing value ratio can be used for evaluating the feature set against the threshold value. The variable having more than the threshold value can be dropped.

$$MVR = \frac{\text{Number of missing values} \times 100}{\text{Total number of observations}}$$

Filter Methods

- Correlation Coefficient
- Variance Threshold
- Mean Absolute Difference (MAD)
- Dispersion Ratio
- Mutual Dependence

Filter vs Wrapper Methods

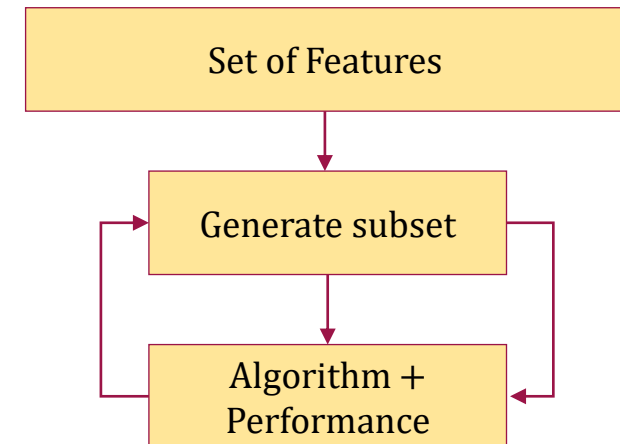
- Wrapper Methods
 - Advantages
 - Achieve **higher recognition accuracy** compared to filter methods since they use the classifier itself in choosing the optimum set of features.
 - Disadvantages
 - Much **slower** compared to filter methods since the classifier must be trained and tested for each candidate feature subset.
 - The optimum feature subset **might not work well** for other classifiers.

Filter vs Wrapper Methods

- Filter Methods
 - Advantages
 - Much **faster** than wrapper methods since the objective function has lower computational requirements.
 - The optimum feature set **might work well** with various classifiers as it is not tied to a specific classifier.
 - Disadvantages
 - Achieve **lower recognition** accuracy compared to wrapper methods.
 - Have a tendency to select **more features** compared to wrapper methods.

Embedded Methods

- Embedded methods combined the advantages of both filter and wrapper methods by considering the interaction of features along with low computational cost. These are fast processing methods similar to the filter method but more accurate than the filter method.
- These methods are also iterative, which evaluates each iteration, and optimally finds the most important features that contribute the most to training in a particular iteration.
- Some techniques of embedded methods are:
 - Regularization
 - Random Forest Importance



Principal Component Analysis

- Principal component analysis, or PCA, is a dimensionality reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set.
- Reducing the number of variables of a data set naturally comes at the expense of accuracy, but the trick in dimensionality reduction is to trade a little accuracy for simplicity. Because smaller data sets are easier to explore and visualize and make analyzing data points much easier and faster for machine learning algorithms without extraneous variables to process.
- So, the idea of PCA is simple — reduce the number of variables of a data set, while preserving as much information as possible.

PCA Algorithm

STEP 1: Standardization

- The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.
- More specifically, the reason why it is critical to perform standardization prior to PCA, is that the latter is quite sensitive regarding the variances of the initial variables. That is, if there are large differences between the ranges of initial variables, those variables with larger ranges will dominate over those with small ranges.

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

STEP 2: Covariance Matrix Computation

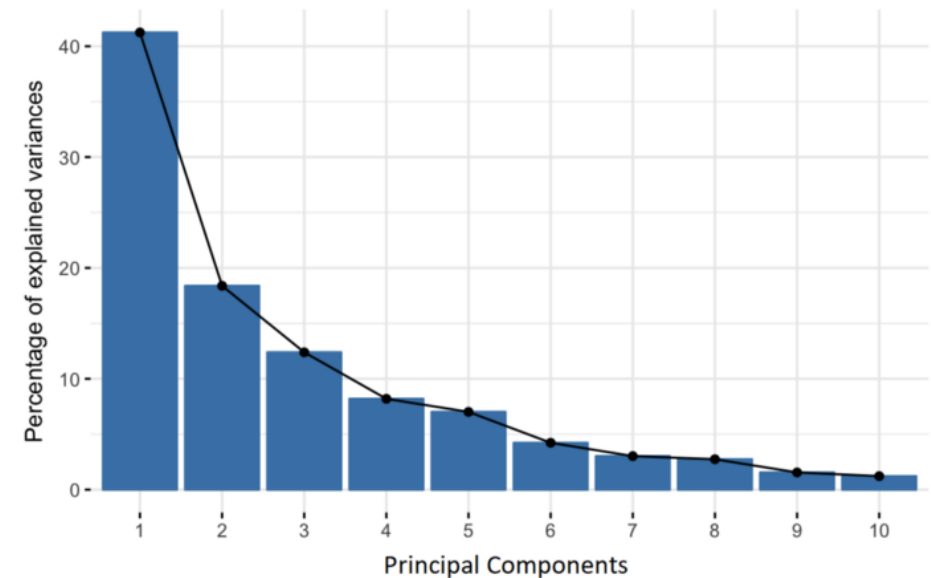
- The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other, or in other words, to see if there is any relationship between them.
- Because sometimes, variables are highly correlated in such a way that they contain redundant information. So, in order to identify these correlations, we compute the covariance matrix.

$$\begin{bmatrix} Cov(x, x) & Cov(x, y) & Cov(x, z) \\ Cov(y, x) & Cov(y, y) & Cov(y, z) \\ Cov(z, x) & Cov(z, y) & Cov(z, z) \end{bmatrix}$$

- Since the covariance of a variable with itself is its variance ($Cov(a, a) = Var(a)$), in the main diagonal (Top left to bottom right) we actually have the variances of each initial variable. And since the covariance is commutative ($Cov(a, b) = Cov(b, a)$), the entries of the covariance matrix are symmetric with respect to the main diagonal, which means that the upper and the lower triangular portions are equal.

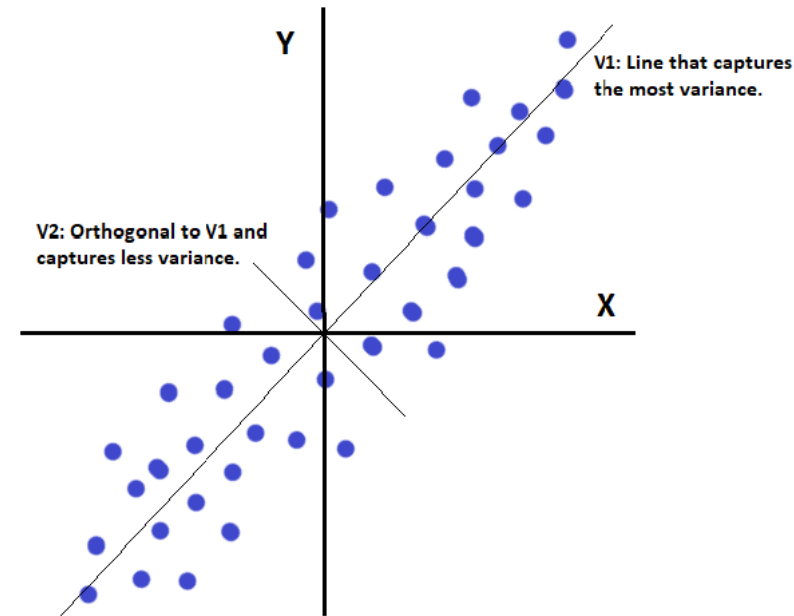
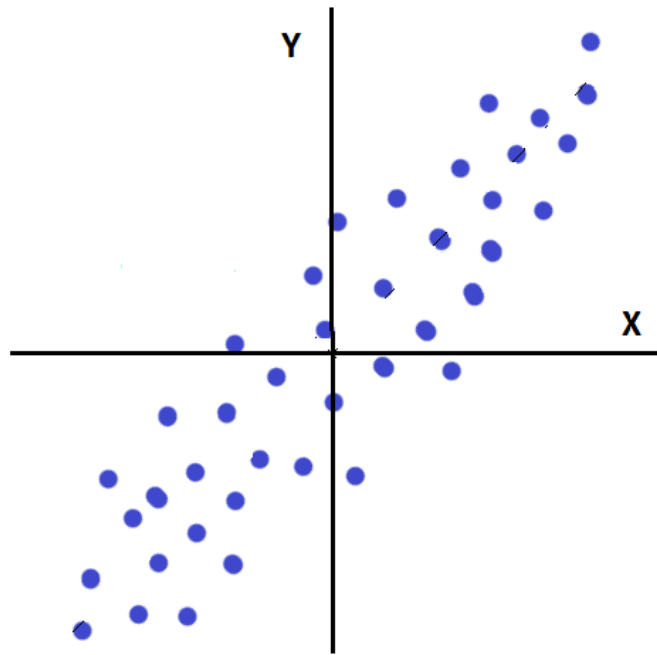
STEP 3: Compute The Eigenvectors and Eigenvalues of The Covariance Matrix To Identify The Principal Components

- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is compressed into the first components.
- So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on.



Geometrically, principal components represent the directions of the data that explain a maximal amount of variance, that is to say, the lines that capture most information of the data.

The relationship between variance and information here, is that, the larger the variance carried by a line, the larger the dispersion of the data points along it, and the larger the dispersion along a line, the more information it has.



- As there are as many principal components as there are variables in the data, principal components are constructed in such a manner that the first principal component accounts for the largest possible variance in the data set.
- The second principal component is calculated in the same way, with the condition that it is uncorrelated with (i.e., perpendicular to) the first principal component and that it accounts for the next highest variance. This continues until a total of p principal components have been calculated, equal to the original number of variables.
- The number of eigenvectors and eigenvalues is equal to the number of dimensions of the data. For example, for a 3-dimensional data set, there are 3 variables, therefore there are 3 eigenvectors with 3 corresponding eigenvalues.
- the eigenvectors of the Covariance matrix are actually the directions of the axes where there is the most variance (most information) and that we call Principal Components. And eigenvalues are simply the coefficients attached to eigenvectors, which give the amount of variance carried in each Principal Component.
- By ranking your eigenvectors in order of their eigenvalues, highest to lowest, you get the principal components in order of significance.

STEP 4: Feature Vector

- In this step, we choose whether to keep all these components or discard those of lesser significance (of low eigenvalues), and form with the remaining ones a matrix of vectors that we call Feature vector.
- So, the feature vector is simply a matrix that has as columns the eigenvectors of the components that we decide to keep.
- This makes it the first step towards dimensionality reduction, because if we choose to keep only p eigenvectors (components) out of n , the final data set will have only p dimensions.

STEP 5: Recast The Data Along The Principal Components Axes

- In this step, the aim is to use the feature vector formed using the eigenvectors of the covariance matrix, to reorient the data from the original axes to the ones represented by the principal components (hence the name Principal Components Analysis).
- This can be done by multiplying the transpose of the original data set by the transpose of the feature vector.

$$FinalDataSet = FeatureVector^T * StandardizedOriginalDataSet^T$$

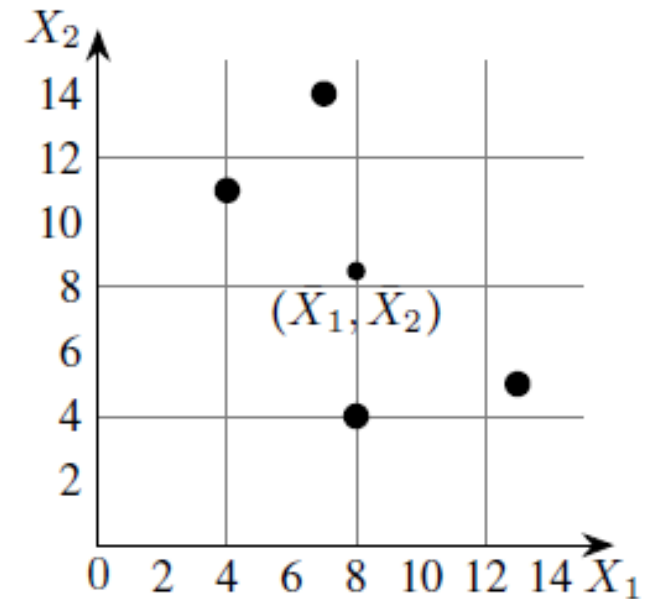
PCA example

Given the data in Table, reduce the dimension from 2 to 1 using the Principal Component Analysis (PCA) algorithm.

	Features	
	X1	X2
Sample 1	4	11
Sample 2	8	4
Sample 3	13	5
Sample 4	7	14

Step 1: Calculate Mean

The figure shows the scatter plot of the given data points.



Calculate the mean of X_1 and X_2 as shown below.

$$\bar{X}_1 = \frac{1}{4}(4 + 8 + 13 + 7) = 8,$$

$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = 8.5.$$

Step 2: Calculation of the covariance matrix.

$$\begin{aligned}\text{Cov}(X_1, X_2) &= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)^2 \\ &= \frac{1}{3} ((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2) \\ &= 14\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_1, X_2) &= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2) \\ &= \frac{1}{3} ((4-8)(11-8.5) + (8-8)(4-8.5) \\ &\quad + (13-8)(5-8.5) + (7-8)(14-8.5)) \\ &= -11\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_2, X_1) &= \text{Cov}(X_1, X_2) \\ &= -11\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_2, X_2) &= \frac{1}{N-1} \sum_{k=1}^N (X_{2k} - \bar{X}_2)^2 \\ &= \frac{1}{3} ((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2) \\ &= 23\end{aligned}$$

So, the covariance matrix is,

$$\begin{aligned}S &= \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix} \\ &= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}\end{aligned}$$

Step 3: Eigenvalues of the covariance matrix.

$$\begin{aligned} 0 &= \det(S - \lambda I) \\ &= \begin{vmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{vmatrix} \\ &= (14 - \lambda)(23 - \lambda) - (-11) \times (-11) \\ &= \lambda^2 - 37\lambda + 201 \end{aligned}$$

Solving the characteristic equation we get,

$$\begin{aligned} \lambda &= \frac{1}{2}(37 \pm \sqrt{565}) \\ &= 30.3849, 6.6151 \\ &= \lambda_1, \lambda_2 \quad (\text{say}) \end{aligned}$$

Step 4: Computation of the eigenvectors

To find the first principal components, we need only compute the eigenvector corresponding to the largest eigenvalue. In the present example, the largest eigenvalue is λ_1 and so we compute the eigenvector corresponding to λ_1 .

The eigenvector corresponding to $\lambda = \lambda_1$ is a vector

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 0 \\ 0 \end{bmatrix} &= (S - \lambda_1 I)X \\ &= \begin{bmatrix} 14 - \lambda_1 & -11 \\ -11 & 23 - \lambda_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \\ &= \begin{bmatrix} (14 - \lambda_1)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda_1)u_2 \end{bmatrix} \end{aligned}$$

This is equivalent to the following two equations:

$$\begin{aligned}(14 - \lambda_1)u_1 - 11u_2 &= 0 \\ -11u_1 + (23 - \lambda_1)u_2 &= 0\end{aligned}$$

Using the theory of systems of linear equations, we note that these equations are not independent and solutions are given by,

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda_1} = t,$$

$$u_1 = 11t, \quad u_2 = (14 - \lambda_1)t,$$

where t is any real number.

Taking $t = 1$, we get an eigenvector corresponding to λ_1 as

$$U_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix}.$$

To find a unit eigenvector, we compute the length of U_1 which is given by,

$$\begin{aligned} \|U_1\| &= \sqrt{11^2 + (14 - \lambda_1)^2} \\ &= \sqrt{11^2 + (14 - 30.3849)^2} \\ &= 19.7348 \end{aligned}$$

Therefore, a unit eigenvector corresponding to λ_1 is

$$\begin{aligned} e_1 &= \begin{bmatrix} 11/\|U_1\| \\ (14 - \lambda_1)/\|U_1\| \end{bmatrix} \\ &= \begin{bmatrix} 11/19.7348 \\ (14 - 30.3849)/19.7348 \end{bmatrix} \\ &= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix} \end{aligned}$$

By carrying out similar computations, the unit eigenvector e_2 corresponding to the eigenvalue $\lambda = \lambda_2$ can be shown to be,

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

Step 5: Computation of first principal components

let,

$$\begin{bmatrix} X_{1k} \\ X_{2k} \end{bmatrix}$$

be the k th sample in the above Table (dataset). The first principal component of this example is given by (here “T” denotes the transpose of the matrix)

$$\begin{aligned} e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} &= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} \\ &= 0.5574(X_{1k} - \bar{X}_1) - 0.8303(X_{2k} - \bar{X}_2). \end{aligned}$$

For example, the first principal component corresponding to the first example

$$\begin{bmatrix} X_{11} \\ X_{21} \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

is calculated as follows:

$$\begin{aligned}
\begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{21} - \bar{X}_2 \end{bmatrix} &= 0.5574(X_{11} - \bar{X}_1) - 0.8303(X_{21} - \bar{X}_2) \\
&= 0.5574(4 - 8) - 0.8303(11 - 8, 5) \\
&= -4.30535
\end{aligned}$$

The results of the calculations are summarised in the below Table.

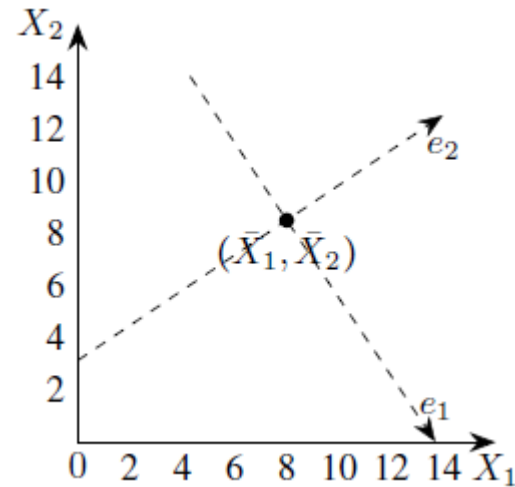
X1	X2	First Principle Components
4	11	-4.3052
8	4	3.7361
13	5	5.6928
7	14	-5.1238

Step 6: Geometrical meaning of first principal components

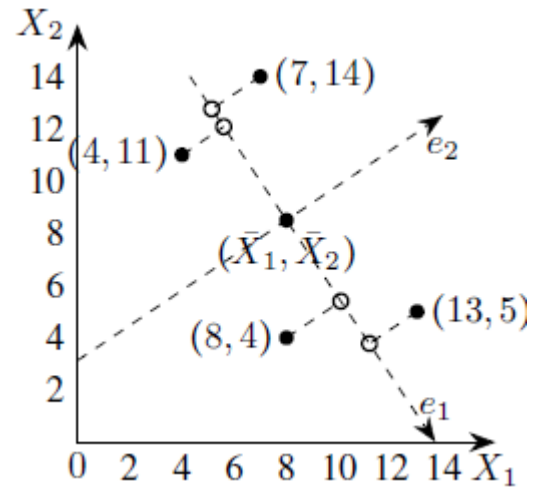
First, we shift the origin to the “center”

$$(\bar{X}_1, \bar{X}_2)$$

and then change the directions of coordinate axes to the directions of the eigenvectors e_1 and e_2 .

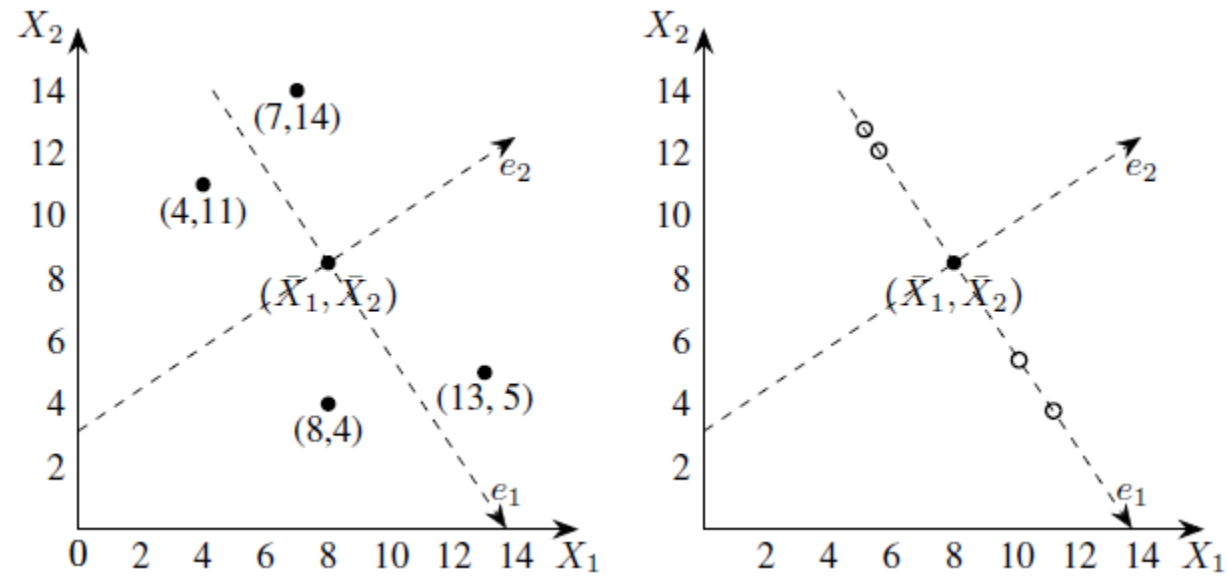


Next, we drop perpendiculars from the given data points to the e_1 -axis



The first principal components are the e_1 -coordinates of the feet of perpendiculars, that is, the projections on the e_1 -axis. The projections of the data points on the e_1 -axis may be taken as approximations of the given data points hence we may replace the given data set with these points.

Now, each of these approximations can be unambiguously specified by a single number, namely, the e_1 -coordinate of approximation. Thus the two-dimensional data set can be represented approximately by the following one-dimensional data set.



Geometrical representation of one-dimensional approximation to the data set

Example:

Solve the following example:

Large size apples (F1)	Rotten apples (F2)	Damaged apples (F3)	Small apples (F4)
1	5	3	1
4	2	6	3
1	4	3	2
4	4	1	1
5	5	2	3

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